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Optimizing Residential Battery Energy Storage Systems Across Frequency Regulation Markets and Energy Arbitrage

Price Forecasting-Driven Battery Scheduling for Participation in Swedish Frequency Containment Reserve Markets

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Abstract

This thesis addresses the optimization of Battery Energy Storage System (BESS) fleets participating simultaneously in Swedish Frequency Containment Reserve (FCR-N, FCR-D Up, FCR-D Down) markets and energy arbitrage. A Mixed-Integer Linear Programming (MILP) model is developed to determine the optimal hourly battery scheduling, considering BESS technical limitations (state of charge, power capacity, efficiency), specific Swedish Frequency Containment Reserve (FCR) market regulations for Limited Energy Resources (LER), and the cost of battery degradation through cycling. As FCR market bids are placed before prices are finalized, the study also develops and evaluates price forecasting models, including ARIMA and simpler baselines, to provide necessary input for the optimization under realistic conditions. Using historical market data from February to December 2024, simulations show that the proposed multi-market optimization strategy significantly increases potential revenue (up to +34.8% under perfect foresight) compared to baseline single-market strategies, such as participating only in Frequency Containment Reserve - Disturbance (FCR-D) markets. Frequency regulation services, particularly FCR-D, are identified as the dominant revenue source, with energy arbitrage offering negligible returns during the study period. Even with imperfect price forecasts, the optimization approach yields revenue gains (+3.7% to +12.0%) over the baseline of only participating in FCR-D, demonstrating practical viability. Furthermore, the analysis indicates that incorporating degradation costs effectively limits battery wear without substantial profit reduction, and heuristic charging constraints can mitigate impacts on the newly introduced household power tariffs of Sweden, while maintaining high net earnings. This research provides a framework for optimally managing residential BESS fleets, enhancing their profitability and contribution to grid stability through intelligent multi-market participation driven by price forecasts.

Keywords

Battery Energy Storage Systems, Energy Storage Optimization, Frequency Regulation, Ancillary Services, FCR, Energy Arbitrage, Peak-Shaving, Multi-Market Participation, Battery Revenue Stacking, Electricity Market Optimization, Battery Fleet Management, Market Price Prediction, Bidding Strategies, Battery Scheduling Algorithms, Linear Programming, Mixed Integer Linear Programming, Virtual Power Plant, Machine Learning, Battery

Dispatch Strategies

Sammanfattning

Rapporten syftar till att optimera användningen av aggregerade batterilagringsystem (BESS) som deltar i olika svenska frekvensregleringsmarknader (FCR-N, FCR-D Upp, FCR-D Ner) samt genomför elpris-arbitrage. En MILP-modell utvecklas för att bestämma optimal schemaläggning av batterierna, med hänsyn till tekniska begränsningar (laddningsnivå, effektkapacitet, verkningsgrad), specifika svenska FCR-regler för Limited Energy Resources (LER), samt kostnaden för batteridegradering genom cykling. Eftersom bud på FCR-marknader lämnas innan priserna fastställs, utvecklas och utvärderas även prisprognosmodeller, inklusive ARIMA och enklare baseline-modeller, för att ge nödvändiga indata till optimeringen under realistiska förhållanden. Genom simuleringar med historisk marknadsdata från februari till december 2024 visar resultaten att den föreslagna multi-marknadsoptimeringsstrategin ökar de potentiella intäkterna (upp till +34,8 % med perfekt prisinformation) jämfört med nuvarande strategier som enbart deltar i FCR-D marknaden. Frekvenregleringstjänster, särskilt FCR-D Ner, identfieras som den dominerande intäktskällan, medans elpris-arbitrage genererar minimala intäkter under den studerade tidsperioden. Även med suboptimala prisprognoser ger optimeringsmetoden ökade intäkter (+3,7 % till +12,0 %) jämfört med enbart deltagande i FCR-D, vilket visar på praktisk användbarhet. Analysen visar även att inkludering av degraderingskostnader i optimeringsmodellen effektivt begränsar batterislitage utan att de totala intäkterna påverkas nämnvärt. Genom att implementera begränsningar i maximal laddningshastighet under olika tider på dygnet kan även effekten på de nyligen introducerade svenska effekt tarifferna minimeras, utan större påverkan på intäkterna. Studien presenterar ett ramverk för optimal styrning av aggregerade bostadsbaserade batterilagringssystem, vilket ökar deras lönsamhet och bidrar till stabilisering av elnätet genom intelligent deltagande på flera frekvenskregleringsmarknader baserat på prisprognoser.

Nyckelord

Batterienergilagringssystem, Optimering av energilagring, Frekvensreglering, stödtjänster, FCR, Energiarbitrage, Effekttoppskapning, Multimarknadsdeltagande, Intäktskombinering för batterier, Elmarknadsoptimering, Hantering av batteriflotta, Prognos av marknadspriser, Budstrategier, Schemaläggningsalgoritmer för batterier, Linjärprogrammering, Mixed Integer Linear Programming, Virtuellt kraftverk, Maskininlärning, Styrstrategier för batterier

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Stockholm, June 2025
Elias Schuhmacher Erik Rosén

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List of acronyms and abbreviations

ARIMA	Autoregressive Integrated Moving Average
BESS	Battery Energy Storage System
BSP	Balance Service Provider
DoD	Depth of Discharge
DP	Dynamic Programming
EOL	End-Of-Life
FCR	Frequency Containment Reserve
FCR-D	Frequency Containment Reserve - Disturbance
FCR-N	Frequency Containment Reserve - Normal
LER	Limited Energy Resources
LP	Linear Programming
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MILP	Mixed-Integer Linear Programming
MSE	Mean Squared Error
NEM	Normal state Energy Management
NPV	Net Present Value
PV	Photo Voltaics
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SoC	State of Charge
SvK	Svenska Kraftnät
TSO	Transmission System Operator

UN	United Nations
VPP	Virtual Power Plant

Chapter 1

Introduction

This chapter describes the specific problem that this thesis addresses, the context of the problem, the goals of this thesis project, and outlines the structure of the thesis.

1.1 Background

With increasing CO_2 levels and temperatures, it is more important than ever that the world moves away from fossil fuels and toward more sustainable sources of energy [1]. Making our energy production more environmentally friendly includes making electricity production rely less on energy sources such as oil, coal and gas and more on energy sources such as wind and solar Photo Voltaics (PV). These energy sources do however have problems, they rely on external factors which are very hard to control, plan, and account for. The reliance on such energy sources means that the energy production is generally less stable which leads to large challenges in ensuring a stable grid. In order for the grid to operate properly, production and consumption must always match. If these do not match, the grid frequency will start to deviate from the frequency it is built for, and with too much deviation the grid may eventually experience significant disturbances or even a total collapse.

In order to balance these increasingly renewable and instable grids, there has been an increasing need for ancillary services which balance the grid. In recent years, Battery Energy Storage Systems (BESSs) have been increasingly used to provide such balancing services. BESSs are very flexible and are able to both absorb energy from and release energy to the grid. The BESSs also have a quick response time, making them even more suitable for the task. Providing these services is compensated financially by the Transmission System Operators (TSOs). Another way to generate economic value with a residential BESS is to engage in energy arbitrage, i.e., charging when electricity prices are low and selling electricity back to the grid (or consuming) when prices are high. Residential BESSs can also be utilized to store self-produced electricity from PV solar production, as well as to reduce peak demand by using stored energy during periods of high consumption. The latter is relevant in the context of power tariffs, which is further explained in the next chapter 2. When it comes to frequency regulation, there is a range of different frequency regulation services that a residential BESS can provide, each with its own set of technical requirements, compensation models, and varying price levels.

With the possibility of using a residential BESS for multiple different revenue streams and frequency regulation services, energy management becomes an increasingly complex task for BESS owners. This work aims to develop models that increase the economic value of a BESS by enabling participation in multiple markets and energy arbitrage, with the goal of optimizing overall profit.

1.2 Problem

Optimizing BESS for multi-market participation is a nontrivial problem for many reasons. The difficulty of the problem lies in choosing what market to participate in at what time, while simultaneously performing energy arbitrage and considering factors such as technical limitations, battery degradation and power tariffs. This requires a multidimensional optimization process with different requirements in all dimensions and where frequency regulation price information is unknown at the time of scheduling. The lack of perfect price information means predictions of compensation prices for the different frequency regulation markets will be needed in order to find an optimal battery schedule.

In addition to price uncertainty, several other factors further complicate the optimization problem. For instance, the State of Charge (SoC) is a continuous decision variable, making the problem hard to discretize and increasing the computational complexity of algorithms that rely on discrete state representations. While Frequency Containment Reserve (FCR) bid sizes are technically discrete, in this work they are assumed to also be continuous, more about this assumption of continuity in section 1.5. Furthermore, the revenue generated under multi-market participation is impacted by external factors such as market prices, which are not known when submitting bids in the auctions. There are also multiple layers of constraints to consider: The BESS itself has technical constraints, Svenska Kraftnät (SvK) has several different constraints for how BESSs are allowed to participate, and the Balance Service Provider (BSP) can have constraints regarding their pre-qualifications.

Lastly, the addition of power tariffs further complicates the problem, since they are based on a combination of the household consumption, which is not known in advance, and the BESS usage profile. In addition, the tariff is not based on aggregate usage but instead on peaks, which means that if BESS is used carelessly, even for only a few hours a month, then the power tariff can be increased significantly.

In summary, the problem of optimizing BESS usage has many aspects that make it challenging, including a continuous state space, lack of price information, technical constraints, and market dynamics.

1.2.1 Research Question

The study aims to answer the following research questions:

1. *How can an aggregated residential battery system be optimally scheduled across multiple revenue streams, including frequency regulation services and electricity price arbitrage, while considering technical and regulatory constraints?*

2. *How well do different methods perform in forecasting short-term frequency regulation prices?*

3. *How does the integration of the market price forecast into the battery optimization model affect the overall economic performance of the BESS operation?*

1.3 Purpose

The desired outcome of the thesis is a method to optimally schedule the battery operation between different revenue streams, while simultaneously adhering to technical limitations and market regulations. In order for the method to be applicable in a real-world scenario, forecasts of the prices for the different frequency regulation markets will be needed, which will be explored in the thesis. By doing this, BESS owners will be able to maximize the economic returns of their investment by automatically scheduling the operation of the BESS based on what is expected to create the most revenue or savings at any given time.

1.3.1 Ethics and Sustainability

In addition to increasing revenue for individual BESS owners, utilizing residential BESS for multiple frequency regulation services and intelligent charging during low-demand hours will simultaneously help stabilize the electric grid, which is important from a socioeconomic and environmental perspective. This will increase the ability to integrate renewable energy sources into the power grid, as production from these resources is often unstable and highly dependent on weather conditions. One potential negative implication of optimizing battery usage for revenue is that it might lead to excessive cycling and faster battery degradation, reducing its lifespan, and potentially leading to increased electronic waste. Therefore, the work in this thesis takes battery degradation into account in order to ensure long-term sustainability.

1.4 Goals

The goals of the thesis can be broken down into three categories: battery scheduling, price prediction for different frequency regulation markets, and evaluation of how these work in combination.

1. **Subgoal 1: Optimal BESS Scheduling**

   The first objective is to develop an optimization method that can schedule the operation of a BESS across multiple services, such as Frequency Containment Reserve - Disturbance (FCR-D), Frequency Containment Reserve - Normal (FCR-N), and local optimization strategies such as peak shaving and price arbitrage. This method should precisely account for the technical limitations of the batteries as well as the technical constraints specific to the Swedish frequency regulation markets. The developed model should be able to use information about electricity spot prices, FCR prices and battery parameters, in order to produce an optimal charging/discharging and frequency regulation participation schedule. Additional factors to consider are battery degradation and limiting power peaks in order to reduce power tariffs.

2. **Subgoal 2: Frequency Regulation Market Price Prediction**

   The second objective is to design and evaluate forecasting models for predicting market prices for the relevant frequency regulation services. This is required because frequency regulation service prices are set in auctions the day before delivery, and prices for the upcoming day are not known at the time when the battery needs to decide on which markets to place bids in. In order to make accurate predictions, different predictive factors will be explored and incorporated into the models. To ensure that the models can be used in a real-world scenario, the models need to accurately consider which information is available at the time of bidding. For the FCR services, bids can be placed in two separate auctions, one of which closes before the spot prices for the upcoming day are made available. In the second auction, spot prices are available as well as procured volume and price from the first auction, which might be used when predicting service prices of the second auction. Different forecasting models of varying complexity will be developed and compared against naive baseline predictions such as using yesterday's prices. The predictions will serve as input to the optimization model.

3. **Subgoal 3: Evaluation of BESS scheduling using sub-optimal price predictions**

   The third objective is to evaluate the expected real-world performance by combing the scheduling algorithm and price prediction models through simulations based on historical market data. To do this, the algorithm will plan the battery operation using only predicted market prices. The actual historical prices will be used to evaluate the performance of the strategy. This will provide insights into how forecast errors impact the revenue potential and whether the combined system would be able to increase revenue in a real world scenario. The evaluation will also consider the perspective of the Virtual Power Plant (VPP) aggregator, in this case the supervising company Greenely. The aggregator shares the revenue from frequency regulation with the BESS owners, and parts of this revenue will be lost when the BESS owners use their battery for price arbitrage instead of frequency regulation. The potential loss in frequency regulation revenue from an aggregators perspective will be quantified and included in the results.

On top of these three more specific goals, this work aims at solving these using a scalable and easy-to-manipulate model. The aim is that battery specifics, market requirements, taxes, etc. should not be hard to change, and it should be as easy as possible to add more markets to the model. The reason for this is to ensure that the work is as useful as possible for as many people/organizations as possible.

1.5 Delimitations

The study has a limited scope, and thus a couple of limitations need to be pointed out.

The study will not consider the impact of PV (solar) production, which is sometimes installed alongside a residential BESS. PV integration could potentially impact the optimal scheduling behavior, as electricity taxes create an incentive to use self-produced electricity instead of buying directly from the grid, and thus PV production predictions may be needed to obtain the optimal battery schedule. In a scenario where multiple residential BESS are aggregated to form a VPP, allowing PV charging for individual BESS leads to challenges with coordination across the VPP fleet. Therefore, to maintain a manageable level of complexity, the analysis assumes no PV integration and treats all batteries as fully controllable assets without local electricity generation.

However, the battery schedule created by the model in this thesis could still be used in scenarios with both PV production and varying household consumption. Whenever the battery is scheduled to discharge, parts of the electricity could go to supplying current household demand, and any excess electricity can be sold back to the grid. Any PV production can be sold directly to the grid, as charging the battery with PV generated electricity will not be possible during hours of active frequency regulation. In this way, the BESS (or the fleet of BESSs) can keep the optimal State of Charge determined by the optimization algorithm at all times. The thesis will not investigate, nor make any claims, regarding the optimality of the battery schedule when PV production and household consumption patterns are taken into account.

The thesis will focus strictly on the Swedish electricity markets, the Swedish frequency regulation markets, and the specific technical requirements from Swedish TSO *Svenska Kraftnät*. The findings may not be applicable or relevant in other countries with different market dynamics and requirements.

The markets explored in this thesis frequently go through regulatory changes and developments, and as such, the assumptions and models used in this thesis are based on the current market structure and regulations as of Q1 2025. Any future changes to the marked design, bidding procedures, or technical requirements may affect the future applicability of the proposed optimization strategies.

When participating in the different markets, there are certain requirements regarding minimum bid sizes and in what increments one can increase a bid; for example, an auction may have a minimum bid size of 1 MW where bidding is only allowed in increments of 0.1 MW. In this thesis such requirements are ignored and it is assumed that one can bid any amount in any market. This will mean some of the bids proposed by the model may not actually reach the minimum bid size and thus are not allowed, similarly, some bids may need to be rounded down to the nearest allowed increment.

Regarding power tariffs, only the Ellevio implementation of tariffs is used. The tariff is calculated by finding the three hours with the highest consumption each month however, at most one hour for any given day will be counted and for consumption between 22:00 and 06:00 only half the consumption counts. The average of these three peaks is then multiplied by a fixed fee of 8.125 €/kW to get the tariff of the month in euros [2].

When calculating the impact the battery on the power tariffs, some simplifying assumptions were made. When looking at what consumption participation in FCR would contribute, only the activations were considered, in reality the battery also charges and discharges using Normal state Energy Management (NEM). It is also assumed that when the battery is discharging, all of it goes to the household, meaning that if the battery has discharged 1 kWh during an hour then the household consumption has also decreased by 1 kWh, this is not necessarily true as the energy can also be sold back to the grid. The calculations also assumes that the household consumption will not be altered with the introduction of power tariffs.

in this thesis, the battery cycle aging is calculated using a linear model, which means charging from 40% SoC to 50% SoC 10 times causes the same amount of battery degradation as charging from 0% SoC to 100% SoC once. This is not entirely accurate, as deep charges and discharges and operation close to the SoC extremes are generally more damaging for the BESS. Instead, using a model to calculate which calculates the equivalent full cycles for all actions using a better model than a linear one may yield more accurate results. For the scope of this work a simpler linear model was deemed good enough since even though the linear model generally overstates the number of equivalent full cycles, the cycle aging was never what led to the scrapping of the battery. All experiments instead showed that if the policy would be extrapolated for the battery lifetime of 10 years (assuming similar usage), then the BESS would still not be close to the 10 000 cycles it is rated for.

This study will only look at the participation in the second FCR auction (D-1). The reason for this is that in the second auction, the spot prices (as well as auction 1 prices) are available, meaning these these can be used for scheduling and improving the predictions accuracy of the second auction prices.

The price of different frequency regulation services have been changing significantly over the last few years. The results in this thesis will be based on historical price data from 2024 (January not included) and should thus be interpreted as the potential revenue that could have been achieved if the model was implemented during that time period. This does not imply that similar revenue levels are guaranteed in future years, as market conditions are highly dynamic and influenced by a range of factors such as regulatory changes, supply and demand for balancing services, and the amount of pre-qualified frequency regulation assets.

1.6 Structure of the thesis

Firstly, chapter 2 starts of by presenting relevant background information about the Swedish energy markets, BESSs, frequency regulation services, as well as energy arbitrage. Secondly, relevant background information regarding computer science and statistical methods is presented. Lastly, related work regarding these different topics is highlighted and discussed.

Chapter 3 presents the steps taken in this work. These steps include the data collection process, assumptions taken, and the construction and evaluation of the model. The chapter also highlights how the different experiments will be evaluated.

Chapter 4 presents results from the multi-market optimization, this includes both revenue, cycles, as well as visualizations of the policy and aggregated policy data. The chapter also presents the results from the price forecasting experiments as well as the combination of multi-market optimization with suboptimal price forecasts. The chapter concludes with presenting different strategies impact on the revenue generated and the power tariff increase caused.

Chapter 5 discussed the results presented in chapter 4, their implications and what may have caused them, as well as considerations when choosing the model.

Chapter 6 presents the conclusions drawn from chapter 4 and 5, the contributions and insights of the thesis, and presents future research directions which have been identified as promising. The chapter and thesis conclude with reflections on how the thesis contributes to a better world by supporting Sustainable Development Goals (SDGs) and ethical considerations.

Chapter 2

Background

This chapter provides basic background information about the Swedish electricity system, frequency regulation services, and residential battery energy storage systems. Additionally, this chapter describes different approaches to BESS scheduling and previous related work in the field.

2.1 Swedish Electricity Markets

The electricity in Sweden is transmitted through a large network of transmission lines that effectively connecting producers and consumers. The process has many different layers to it, making it a complex yet essential infrastructural system that ensures a reliable supply of electricity to households, industries, and businesses. The transmission system is divided into different distribution grids: the main, regional, and local grids [3]. The main grid consists of high voltage transmission lines (400 or 220 kV) that carry electricity across large distances, connecting the southern parts of Sweden with the northern parts, where most of the electricity is produced [4]. The main transmission lines also expand across the nation's borders, connecting Sweden with Denmark, Norway, Finland, Germany, Poland, and Lithuania [3]. The main transmission grid in Sweden is managed by the state-owned TSO SvK, while the smaller regional and local grids are owned and managed by private distribution system operators such as Ellevio and Vattenfall.

Sweden and the rest of Europe have deregulated electricity markets, which means electricity is bought and sold on open markets where prices are determined by supply and demand [5]. The market is divided into numerous submarkets with different time frames and purposes. In Sweden and the rest of the Nordics, trading occurs primarily on Nord Pool, the leading power market in Europe [6].

2.1.1 Forward Market

The market with the longest time frame is the forward market, which allows participants to trade electricity contracts up to 10 years in advance of the actual delivery of electricity. These contracts are usually financial instruments used to hedge against future price volatility, rather than physical commitments to deliver electricity.

2.1.2 Day-ahead market

The day ahead market is where most of the electricity volume is traded and the hourly electricity *spot prices* for the upcoming day are set [7]. Producers and large-scale consumers place bids and offers for hourly intervals in an auction that closes at 12:00 the day before planned delivery. The electricity prices for each hour of the following day are therefore determined by supply and demand and will be available between 12 and 36 hours in advance, shortly after the auction has closed. When the auction for the upcoming day closes, the next auction for the day after tomorrow opens, resulting in a 24 hour bidding period for each day.

The day ahead market is divided between Sweden's four electricity price areas: SE1, SE2, SE3 and SE4. This means different areas have different hourly spot prices, where the northern areas SE1 and SE2 typically exhibit lower electricity prices compared to the middle (SE3) and southern (SE4) Sweden [4]. This dynamic is caused by the limited transmission capacity between zones, and most of the electricity is produced in northern Sweden, while consumption is higher in the more densely populated southern areas [4].

2.1.3 Intraday market

While most of the electricity volume is traded on the day-ahead market [7], the intraday market complements the day-ahead market by allowing participants to adjust their supply and demand positions up to one hour before delivery. This can happen due to unexpected changes in either generation or consumption forecasts, which is becoming increasingly frequent as more weather-dependent electricity sources are integrated into energy production [8][9]. The intraday market operates continuously and helps to stabilize the grid by reducing last-minute mismatches between supply and demand.

2.1.4 Upcoming changes to electricity markets

One of the major changes planned for 2025 is the switch from hourly intervals in the day-ahead market to 15-minute intervals. This means that there can be four different *spot prices* for electricity during the same hour, where before the electricity price was kept constant for the whole hour. The main reason for this is to be able to handle the increased variability in production introduced by the transition towards more sustainable, weather-dependent energy sources [10].

2.1.5 Power Tariffs

The electric grid has limited capacity, and if too much electricity is required at the same time, the grid risks not being able to supply the required amount. One solution is to expand existing capacity by investing in new transmission lines and upgrading the distribution infrastructure. However, these upgrades are both costly and time-consuming. Therefore, another approach is to utilize the existing grid more effectively, by shifting electricity consumption to periods when demand is lower, and by reducing peak-loads. To incentivize this, Swedish *Elmarknadsinspectionen* decided 2022 that electricity grid companies will have to base parts of the electricity grid fee on the users peak power consumption of the month, this is a power tariff. [11]

Exactly how the power tariffs are implemented varies between electricity companies. One implementation is to look at the hour with the highest average power-consumption and charge a fixed amount multiplied with this peak-hour consumption. Other companies look at the three highest peaks of the month and consider the average, a model like this is shown in Figure 2.1.

In order for customers to reduce their power tariff, they would have to even out their consumption pattern by avoiding simultaneous use of high-power appliances, such as electric vehicle chargers, ovens, and washing machines. The power tariffs may also vary depending on the time the peak occurred. As an example, the Swedish Energy company Ellevio only considers half of the power peak if it occurs at night, between 22:00 and 06:00 [2]. Other electricity companies only consider power peaks during the daytime and weekdays, meaning weekend and night consumption will not affect the power tariff at all. There may also be differences between seasons and time of year, with companies charging a higher fee per kW during winter months, when the grid load is typically higher. Note that since the power tariff utilizes the average power used in an hour, which is calculated as $Average\,power\,(kW) =$

$\frac{Energy\ use\ (kWh)}{Time\ (h)}$, and the timestep is one hour, it simplifies to $Averagepower = \frac{Energy\ use}{1} = Energy\ use$. Therefore, it is equivalent to look at the average power consumption and the energy consumption of the hour when calculating the power tariff even though these have different units and are different physical quantities.

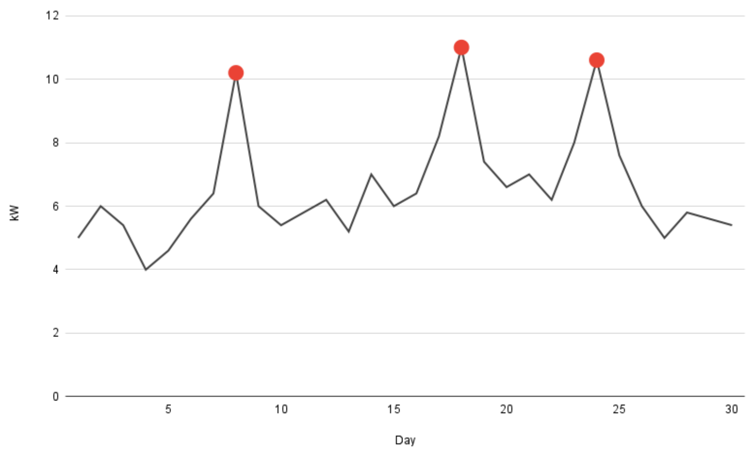


Figure 2.1: Illustration of power-based tariffs.

As a consumer, it becomes important to get acquainted with the specific tariff structure of the electricity provider and plan consumption accordingly. By shifting consumption patterns, such as charging electric vehicles at night, significant savings can be generated.

2.2 Battery Energy Storage Systems

A BESS is a term used to describe a setup that includes batteries and supporting infrastructure, such as inverters, that can store and manage electricity. The role of BESS has become increasingly crucial as the energy landscape shifts towards more renewable energy, where excess electricity generated by intermittent sources such as wind and solar must be stored for later use [9]. Recent advancements in battery technology, such as improvements in lithium-ion batteries, have made BESS more efficient and cost-effective, making it more economically viable for both large-scale applications and residential use [12].

Individual homeowners can choose to install a residential BESS for various reasons. One reason is to store surplus electricity from installed solar panels, and thus increase self-consumption and dependency on the grid. A residential BESS can also be used to store electricity when energy prices are low, and later consume or sell the stored energy when prices are high. Additionaly, the ability to quickly switch between charging and discharging makes BESS effective in providing *frequency regulation* to the grid [9].

2.2.1 Battery Degradation

BESS are limited in the sense that they get worse and worse over time, this phenomenon is generally called degradation. For BESS there are two main sources of degradation, these are calendar aging and cycling degradation [13][14]. Calendar aging is when the battery performance is lowered due to battery age, this is caused by a few different chemical processes such as electrolyte decomposition and so-called solid–electrolyte interphase layer growth [14]. These processes build up over time and slowly reduces the performance of the batteries. Cycling degradation or cycle aging is a combination of several different chemical processes such as the same solid–electrolyte interphase layer growth as seen in calendar aging and particle fracture. These processes are affected by factors such as the load profile (how the battery is charged and discharged), the temperature, and the SoC [14].

2.3 Frequency regulation

For the electric grid to operate properly, there must always be a balance between supply and demand [15][16]. The Swedish grid must be kept at a grid frequency of 50 Hz, however, differences in supply and demand result in deviations from 50 Hz [15][16]. Because of this, the TSO, SvK, needs to ensure that the grid can always be balanced, which is done by frequency regulation. SvK has six different types of stabilizing resources, all of which work by changing their production or consumption of energy to move the grid toward a balance between supply and demand. These different types of reserves have different functions and different activation criteria, such as frequency deviation and activation speed [16]. SvK buys these different services from a set of external actors. In order to sell balancing services to SvK, you need to be an approved BSP [17]. These services are bought in auctions hosted by SvK and the volume being bought for each service is set by SvK based on various criteria created to ensure that the frequency can always be restored.

2.3.1 FCR-D

FCR-D is a set of frequency regulation services whose purpose is to restore the grid frequency in the event of a significant disturbance. FCR-D is split up into FCR-D Up and FCR-D Down, whose purposes are to increase and decrease the grid frequency in the event of an disturbance, respectively. FCR-D Up is activated if the grid frequency $F \in [49.5, 49.9]$ where the activation scales linearly with the deviation [18]. Similarly, FCR-D Down is activated if the grid frequency $F \in [50.1, 50.5]$ where the activation scales linearly with the deviation [19]. The activation size based on the frequency can be seen in Figure 2.2.

The FCR-D services are both bought hourly in their own separate auctions held on the day before the delivery hour. The first auction closes at 00:30 the day before planned delivery, which is before spot prices are set, and the later, complementary auctions close at 18:00 o'clock, which is after the spot prices have been set and made available [18][19]. Both have their prices set using marginal pricing, meaning that all accepted bids will be paid the same amount as the most expensive accepted bid [18][19] . The price of these services have decreased significantly during 2024, at least partially because of an increase in supply as BSPs have had a significant increase in the total pre-qualified volume of these services. [20][21][22][23][24].

In order to participate in FCR-D, SvK requires the resource to be able activate fully for 20 minutes [25]. In addition, for LER-resources SvK requires 20% of bidded capacity to be reserved in the opposite direction [25]. As stated previously, FCR-D is only activated during significant disturbances, which are quite rare, also activation of FCR-D does not necessarily mean it is activated fully. This means that on average FCR-D has an average activation of less than 0.1% [24].

2.3.2 FCR-N

Frequency Containment Reserve - Normal or FCR-N is a frequency regulation service whose purpose is to adjust the grid frequency towards 50 Hz during normal operations. FCR-N is activated when the grid frequency is $F \in [49.9, 50.1]$ and the activation linearly scales with the deviation from 50 Hz [26]. The activation size based on the frequency can be seen in Figure 2.2.

FCR-N is bought hourly at auctions the day before the timeslot. The first auction is held 00:30 o'clock, which is before the spot prices are set, and the later, complementary auction is held 18:00 o'clock, which is after the sport-prices have been set [26]. The price for FCR-N is set using marginal pricing, meaning that all accepted bids will be paid the same amount as the highest accepted bid [26]. In addition to this, the FCR-N provider is paid based on how much the resource has been activated during the time period, called energy activation compensation [26]. On average, the activation compensation is about 3.5% of the total revenue when doing FCR-N [27]. This means the activation compensation is about $\frac{1}{1-0.035} - 1 \approx 0.036$ of the capacity prices on average.

In order to participate in FCR-N, SvK requires the resource to be able to fully activate for one hour in both directions [25]. In addition, for Limited Energy Resources (LER) resources SvK requires 34% of bidded capacity to be reserved in both directions [25]. FCR-N resources have an average activation of 15% [24].

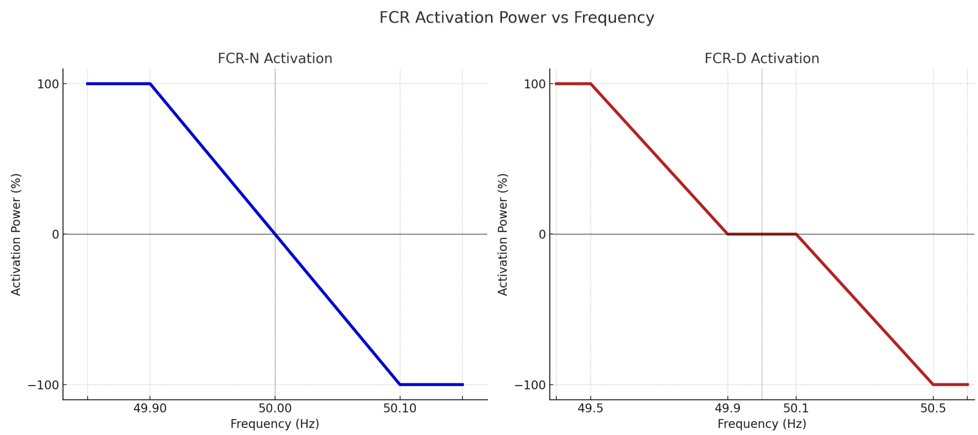


Figure 2.2: FCR activation power as a function of grid frequency. Right side of FCR-D graph corresponding to FCR-D Down and left side corresponding to FCR-D Up

2.3.3 Limited Energy Resources

A resource is classified as a Limited Energy Resources in the FCR markets if its energy reservoir is insufficient to sustain two hours of continuous FCR delivery at the maximum pre-qualified capacity [28]. This typically includes battery energy storage systems. To ensure that the LER systems can provide frequency regulation while keeping their state of charge within required levels, specific rules have been introduced for LER resources in terms of endurance, energy recovery, and capacity reservation.

When participating in FCR-D, LER resources are required to be able to sustain 20 minutes of full activation. For FCR-N, it is instead required to be able to sustain full activation for 60 minutes in both directions [28]. For example, when it comes to FCR-N, the limiting factor for 1-C batteries is the amount of stored electricity, not the power capacity. This is due to the fact that FCR-N requires the BESS to be able to activate 100% of the bid capacity for a minimum duration of 1 hour [28]. A 10 kW / 10 kWh BESS at 50% SoC could therefore only bid a maximum of 5 kW in FCR-N, since the BESS would otherwise run out of electricity if there is a full activation for the whole hour. When it comes to FCR-D, the minimum required activation time is 20 minutes [28], which leads to the power capacity becoming the limiting factor instead of the storage capacity.

Since a BESS only has a limited amount of energy available, its SoC must somehow be restored if participating in FCR for consecutive hours. This is done using NEM which is a way for the battery to be able to restore its SoC while still participating in FCR. There is also a more drastic version called Alert state Energy Management, which means that one is required to tell SvK if the resource has a critical SoC and may not be able to contribute to FCR. Since NEM means the resource may sometimes be charging or discharging, this means the reference effect may actually not be what was initially planned. To account for this, SvK limits the bid size for LER so that when participating in FCR-D 20% of the bidded capacity must be reserved in the opposite direction, and when participating in FCR-N, 34% of the bidded capacity needs to be reserved in both directions [28]. The requirements from SvK regarding reserving 20% power in the opposite direction means that the actual bid sizes are limited to 8.3 kW in both directions for FCR-D when participating in both markets simultaneously [28].

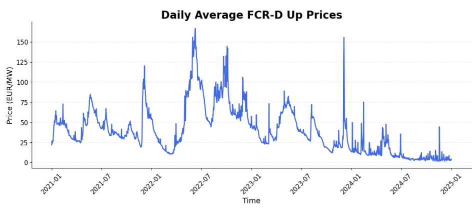
2.3.4 Frequency Regulation Pricing

There are two forms of compensation when selling frequency regulation services: capacity and activation. Capacity payments compensate providers for having a certain capacity ready for frequency regulation. Whether the resources are activated or not, i.e when a disturbance occurs, is not taken into consideration. Similar to an insurance, actors are compensated solely for their readiness to stabilize the grid when needed.

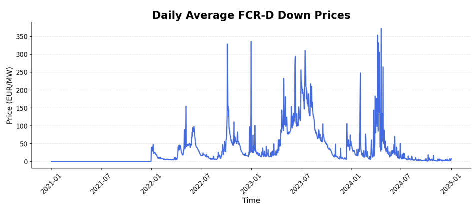
Activation payments, on the other hand, are based on the amount of actual energy supplied during a frequency deviation. The prices for the activated energy are taken from the mFFR market [16]. The net sum of activated energy during the delivery hour determines whether the up-regulation or the down-regulation price will be used. Whether actors are compensated for capacity or activation varies between frequency regulation markets. The FCR-D market provides only capacity compensation [19], while the FCR-N market provides both capacity and activation compensation [26].

In the last years, especially 2024, prices for FCR-D Up, FCR-D Down and FCR-N have been declining sharply, as can be seen in Figures 2.3 and 2.4. This is in part due to an increase in the supply of energy storage systems and wind power plants pre-qualified to provide ancillary services [20][29]. At the beginning of 2023, the pre-qualified volume of FCR-D resources coming from energy storage systems amounted to less than 20 MW [29]. In 2024, the amount had increased to 140 MW, and in the beginning of 2025, the reported volume increased to 1 180 MW [29].

One possible explanation is that the previous high prices of FCR-D have attracted a lot of investments in Battery Energy Storage Systems. As an example, previous studies based on the historically high prices of 2022 have suggested a short payback time of 1-2 years for batteries participating in FCR-D, which made the investment highly attractive [30]. Unfortunately for investors, the increase in supply led to significantly lower market prices, with the average price in December 2024 being less than a tenth of the average prices observed in 2022 and 2023 [31]. This has led actors to look for alternative revenue streams for their BESS, such as participating in different ancillary markets, engaging in price arbitrage, or utilizing the BESS for increased solar self-consumption [32].



(a) FCR-D Up Prices over time.



(b) FCR-D Down Prices over time.

Figure 2.3: FCR-D Prices for Up and Down regulation.

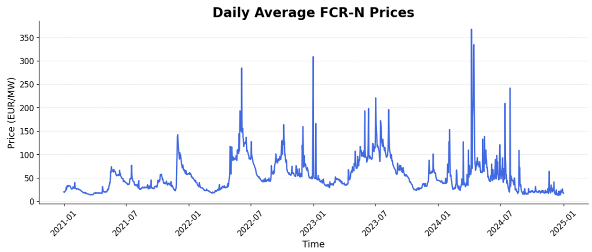


Figure 2.4: FCR-N Prices over time.

2.3.5 VPP Technology

When providing ancillary services to the electric grid, limited resources such as a single residential BESS, are too small to generate any meaningful impact on the electric grid. Because of this, frequency regulation markets have a minimum bid size. For FCR-D and FCR-N, the minimum bid size is 0.1 MW in Sweden [33]. A typical residential BESS has a power capacity of around 10 kW (0.01 MW) [9], which in itself would be far below the minimum bid size. This is where the use of VPP technology comes into play.

Virtual Power Plants refer to a system of distributed energy resources that are aggregated and controlled as a single entity. These resources can be spread out geographically and typically include assets such as rooftop solar panels, residential batteries, and electric vehicle chargers [34]. Using software and communication technologies such as Wi-Fi and Bluetooth, the operation of each single entity can be synced with all the other entities in the VPP, thus being able to generate a larger power output and impact on the grid than any single resource alone.

The practice requires a so-called *aggregator*, an entity responsible for coordinating and managing the distributed resources within the VPP. The aggregator acts as an intermediary between resource owners and the electricity markets or grid operators. This allows home owners with a small residential BESS to connect their battery to an aggregator, which can utilize their resource in combination with other similar resources to sell frequency regulating services to the electric grid operators. In doing so, individual residential BESS owners can generate income doing frequency regulation even if their resource is far below the minimum bid capacity required by the markets. The revenue generated is typically shared between the aggregator and the resource owners. In Sweden, example aggregators include companies such as CheckWatt, Tibber and Greenely.

There are different ways to utilize VPP technology to help balance the grid. One way is to control consumer consumption in a synchronized manner, such as by reducing the charging speed of electric vehicles in multiple households simultaneously when the grid load is high. Traditionally, this way of controlling consumer consumption was the main use of VPPs [34]. As battery technology evolved and solar power became more widespread, VPPs are now increasingly used to both consume and supply electricity back to the grid, allowing them to contribute to frequency regulation in multiple different ancillary service markets.

2.4 Energy arbitrage

The idea behind energy arbitrage is to buy energy when it is cheap, store it, and then sell it when the price is high. The amount of money one can make on price arbitrage is limited by several factors, including the energy storage capacity, price volatility, as well as the round-trip efficiency of the energy storage system [35].

Another factor impacting the profitability of energy arbitrage is electricity taxes, VAT, and any additional grid fees. This results in a difference in the price paid for electricity compared to the price received when selling electricity back to the grid. As of 2024, the electricity tax in Sweden is 0.428 SEK/kWh [36].

2.5 Computer Science and Statistical methods

In order to fulfill the goals of this work, i.e. find the optimal battery scheduling, perform price prediction, and evaluate BESS scheduling with imperfect price information, some technical methods will be needed. There are several different approaches to the problems, some of which will be discussed in Section 2.6. This section introduces the computer science and statistical techniques used in this work, providing the relevant background for the methods models that have been built.

2.5.1 ARIMA

Autoregressive Integrated Moving Average (ARIMA) is a statistical forecasting method [37]. ARIMA consists of three main parts, auto-regressive (AR), integrated (I), and moving average (MA). **AR** means future values are predicted using historic values, **I** means the model contains an integration step in order to make the time series stationary (have constant mean, variance, and auto-correlation), and **MA** means a moving average of previous observations is used. ARIMA models are commonly denoted as $ARIMA(p, d, q)$ where:

- p is the order of the AR part.
- d is the degree of differencing used to transform the time series towards being stationary.
- q is the order of the MA part.

ARIMA works especially well with lagged linear relations and for non-stationary time-series. [37]

2.5.2 Prophet

Prophet is an open-source forecasting tool developed by Meta that is designed to handle time series data that exhibit seasonality and a trend component [38]. It is based on a model in which the forecast $y(t)$ is decomposed into trend $g(t)$, seasonality $s(t)$, and holiday effects $h(t)$:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

Where ϵ_t represents the error term. The trend component $g(t)$ can be modeled either as a piecewise linear function or as a logistic growth curve, with automatic detection of change points. The seasonality component $s(t)$ is modeled using a Fourier series, allowing the model to capture flexible periodic patterns (e.g., weekly or yearly). Holiday effects $h(t)$ are modeled using indicator variables for known recurring events. Priors are assumed for the different components and they are then optimized using bayesian optimization to find the maximum a posteriori estimation.

Prophet is particularly well suited for business time-series with multiple seasonalities, missing data, and outliers [38]. One of Prophet's advantages is that it requires minimal manual tuning and can automatically detect change points in the trend.

Additionally, the Prophet model can also incorporate user-defined *external regressors* (also known as input features or exogenous variables) to hopefully increase the prediction accuracy. The extra input features will be used both when training the model and when producing predictions. Examples of possible regressors include weather data like temperature or rainfall.

2.5.3 Markov Decision Process

A Markov Decision Process is a class of sequential decision processes. They are defined by a set of possible states, possible actions in each state, rewards (or costs), transition probabilities, timesteps, time horizon, and the optimality criteria [39]. In order for the process to be considered Markov, the rewards and transition probabilities only relies on the current state and action, not on the entire history of the process. A set containing what action to take depending on the current state is called a policy, and the set containing the best action for every state is called the optimal policy. There are several different algorithms used to finding the optimal or close to optimal policy of an Markov Decision Process which have different use-cases depending on how the Markov Decision Process is structured as well as how much information is known.

2.5.4 Dynamic Programming

Dynamic programming is a theoretical framework for studying multi-stage processes and finding sequences of operations, such as finding an optimal policy for a Markov Decision Process in a recursive manner [40][41]. It is also known as backward induction as it starts with the end-states of the sequence and then runs backward filling in the values of the previous states, because of this dynamic programming is only applicable if there are end-states [41].

2.5.5 Linear Programming

Linear programming is a method for optimizing linear objective functions with a set of linear constraints [42]. A general linear program can be written as:

$$\begin{aligned} \text{Maximize} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{Subject to} \quad & A\mathbf{x} \leq \mathbf{b} \\ \text{and} \quad & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

In this characteristic equation $\mathbf{x}$ is the vector of the decision variables , $\mathbf{c}^\top$ is the objective coefficients vector, $\mathbf{A}$ is the constraint matrix and $\mathbf{b}$ is the constraint bounds vector [42]. The method has many different algorithms in order to do the actual optimization, and common algorithms include the Simplex method which in practice runs in polynomial time, however, in the worst case scenario it can take exponential time to solve. Linear programming methods are guaranteed to find the optimal solution as long as the program satisfies a few conditions [42].

2.5.6 Mixed Integer Linear Programming

Mixed integer linear programming is an extension of linear programming which also allows decision variables to be constrained to integer variables [43]. A general Mixed-Integer Linear Programming (MILP) program can be written as:

$$\begin{aligned} \text{Maximize} \quad & \mathbf{c}^{\top}\mathbf{x} \\ \text{Subject to} \quad & A\mathbf{x} \leq \mathbf{b}, \\ & \mathbf{x}_i \in \mathbb{Z} \quad \text{for } i \in I, \\ & \mathbf{x}_j \in \mathbb{R} \quad \text{for } j \notin I. \end{aligned}$$

Where **I** is the index set of variables constrained to be integers. The addition of integer constraints makes the programs significantly harder to solve compared to Linear Programming (LP) because of the combinatorial nature of discrete constraints. There are however solvers which efficiently explore and narrow down the solution space and find the optimal solution [44].

2.6 Related Work

2.6.1 Battery Scheduling for Frequency Regulation Services

Several studies have explored optimal scheduling of BESS across multiple electricity markets. In [30], the potential revenue of a BESS participating in different Swedish Frequency Regulation markets was explored and compared. The calculations, which are based on 2022 price data, suggest that participating in the FCR-D Up market yields the highest revenue for a BESS. The study further explores the potential revenue of dynamically selecting the market with the highest price each hour, which leads to higher profits. However, the study does not consider the possibility of participating in multiple frequency regulation markets simultaneously and splitting the battery capacity between multiple markets during the same delivery hour. For example, as mentioned in one of the interviews in the same study, participating in both FCR-D Up and FCR-D Down at the same time is possible for batteries that have a SoC close to 50%, which allows the battery to both start charging and discharging, thus regulating the frequency in both directions.

Another study [45], accounts for this possibility of bidding simultaneously in FCR-D Up & Down. The study concludes that bidding equally in FCR-D Up & Down for the same hours is the most profitable option overall, but it also proposes that in some rare cases when the price difference between FCR-D Up and FCR-D Down is large, it can be be beneficial to bid only in one of these markets, as that would allow a slightly higher bid capacity than what is possible when bidding in both markets simultaneously [28].

In [46], different control mechanisms to recover the State of Charge (SoC) of a battery participating in the FCR-N market were explored. Strategies for recovering the SoC to an accepted level are needed to meet the market requirements of being able to supply full capacity for a set minimum period of time. The study proposes three methods, first of which is to charge/discharge in the dead-band range, i.e., when the frequency is very close to 50 Hz and no activation is required. A similar approach is used in [47][48].

As of 2019, this approach is no longer allowed in the Nordic frequency region, as grid operators fear this can cause disturbances if batteries start to regulate their SoC as soon as the frequency returns to 50 Hz [46]. Therefore, a second proposed strategy is to reserve time between bids where no bids are made in order to recover SoC for the next bidding period. Examples of this strategy can be found in studies from 2019-2022 e.g. [30][49], where this approach is used to bid only every second hour in the frequency regulation markets. During this time period, recovery requirements by SvK prohibited bidding in the frequency regulation markets for two consecutive hours with the same LER resource [28].

A third approach proposed by the authors is to reserve some of the battery capacity to regulate SoC during delivery hours. This means that resources do not bid all of their available capacity, so that even if frequency regulation requires the resource to activate 100% of the bidded capacity, some capacity remains available to manage the SoC. As of 1 September 2023, this approach is favored by SvK, and bidding for all hours of the day is now allowed, both for FCR-N and FCR-D, provided LER resources reserve some of their capacity to regulate the SoC [28]. Examples of this approach can be found in [45][50].

As shown in [50], the new requirements for limited energy resources set by SvK give rise to some interesting possible bid combinations of FCR-D Up, FCR-D Down & FCR-N. The authors of the study start with precise mathematical formulations of the technical requirements for the different markets, and then proceed to use a MILP approach to find the optimal combination of bids and bid sizes based on prices from 2022. As some of the other studies have shown, the most frequent bid option given by the optimal solution is the combination of both FCR-D Up & FCR-Down simultaneously.

Their study also confirms the results from [45], that sometimes bidding in only one of FCR-D Up or FCR-D Down can be optimal when there is a significant price difference between the two. What the study also shows is that the technical requirements allow for splitting the battery capacity and bid in not only FCR-D Up & Down, but also in FCR-N at the same time. According to their findings, it was almost always better to combine bids in the FCR-N market with bids in the FCR-D market, compared to only bidding in the FCR-N market.

2.6.2 Energy Arbitrage

[51] implemented MILP in order to optimally schedule battery usage for price arbitrage. The optimization considered several battery limitation factors; however, battery degradation was not considered. The study found that compared to other markets, the Swedish market was not as suitable for price arbitrage and that a battery performing optimal arbitrage uses about 500 cycles per year. Similarly, [52] found that the Swedish market is not well suited for arbitrage due to low price fluctuations. For the 22 European countries studied, the Net Present Value (NPV) of using a battery simply for energy arbitrage was found to be negative. This study found that the optimal strategy uses between 57 and 257 cycles per year between 2016 and 2022, significantly less than the 500 found in [51]. In contrast to these approaches, [53] models cycle degradation and calendar degradation and then evaluates how different strategies affect the lifetime and lifetime revenue of the battery. The study found that more aggressive approaches, in which the battery is charged and discharged quickly, led to shorter battery lifetimes but to higher lifetime profitability. However, one limitation of the study was that it locked the battery's charge/discharge rate to a specific value, meaning the battery was only able to charge/discharge at that rate for an entire hour. This means the obtained policy can not dynamically choose to sacrifice battery health for profits in case of a very lucrative arbitrage opportunity and then use a more modest charge rate in order to make small profits with low amounts of degradation.

2.6.3 Battery Degradation

What sets many of the previous studies apart is how battery degradation is considered in the calculations. When it comes to optimizing between FCR-D and FCR-N, a key difference is that FCR-N requires more activation than FCR-D, leading to increased battery cycle degradation [45]. This is because FCR-N works continuously under normal conditions to bring the grid frequency back to 50 Hz, while FCR-D only activates once there is a significant frequency deviation ($\pm$0.1 Hz). Additionally, engaging in energy arbitrage will cause increased battery degradation as it relies on large charge and discharge actions. In some studies [45][51], battery degradation is mentioned but neglected in calculations. In [46][49][54][55], an algorithm called the rainflow-algorithm is used to calculate the number of full battery cycles from a varying SoC profile. Studies utilize this cycle count in different ways.

In [30], the cycle cost is determined by a simple division of the purchase price of a battery and the number of cycles it is rated for. For cycles with a smaller Depth of Discharge (DoD) than 100%, a fraction of this cycle cost is used instead. This approach is deemed too simplistic by [56], where the authors point to the fact that cycles with a smaller DoD do not cause as much battery degradation per energy output as cycles with a higher DoD. Consequently, by only doing small charge/discharge cycles, the battery could potentially produce a higher total amount of electricity during its lifetime. The authors propose a way to calculate the marginal cost of using battery cycles of varying DoD by using a near-quadratic cycle depth stress function, which accounts for this fact. In order to solve the optimization problem using the non-linear cycle depth stress function, piecewise linearization is used. A similar approach can be found in [50][46][57].

In [50] and [46], the authors provide another level of detail in the battery degradation model, which models calendar aging caused by different levels of SoC. This is based on the idea that batteries experience different degrees of calendar aging at different levels of SoC. By using lab-tested battery degradation functions for both calendar and cycle aging given by the battery manufacturers, degradation is included in the objective function of the optimization problem. In [50], the authors suggest that by including degradation in the optimization objective, battery degradation can be decreased by 5-29 %, without significantly impacting revenue.

2.6.4 Peak Shaving

In [58] a model was built that aims to find the optimal peak shaving strategy. The model works by iteratively searching over different spans of shave levels in order to find the most suitable one. The model is then analyzed statistically and it is found that one can essentially input a shave level, and then one can find the probability of success for the algorithm. Another interesting finding on peak shaving is that the potential profitability of BESS can be larger in the case that there is a power tariff [49].

2.6.5 Multi-Objective Optimization

In addition to the frequency regulation market optimizations, there have been several studies aiming at optimizing BESS usage across different revenue streams. One such example is the one proposed by [49] which optimizes between peak shaving and frequency regulation. It does this by employing a simple rule-based strategy; do frequency regulation during the night and peak shaving during the day. This strategy was found to decrease the payback time of 17.9 years (using only peak shaving) to 7.8 years. However, this method does not find, or even attempt to find, an optimal policy. It also does not optimize between different revenue streams during the night; instead, only one service is chosen and evaluated. Another, more advanced approach is done in [59] in which battery usage is optimized to:

- Increase self-consumption of PV
- Peak Shaving
- Price Arbitrage

The algorithm does this by forecasting PV production and power consumption and then finding optimal strategies assuming these forecasts are correct. Forecasting is done using artificial neural networks, Long Short-Term Memory (LSTM), as well as a baseline that takes the prices of the last week. Then a sensitivity analysis is conducted using a Monte Carlo method. The study found that LSTM forecasting worked the best, however, with current market dynamics, even with this algorithm the batteries had a negative NPV. In [60] a genetic algorithm is used to optimize systems for energy arbitrage and FCR-services. However, the study does focus more on optimizing the physical setup, such as choosing battery specifications, combining with PV etc. rather than finding an optimal policy for a battery of a certain size.

2.6.6 Linear Programming approaches

A common approach to battery optimization found in previous studies is linear programming. Linear programming methods formulate the battery optimization problem as a mathematical model with linear objective functions and linear constraints. The mathematically formulated constraints are based on the technical constraints of the BESS, as well as the different market requirements. These optimization problems can be solved by using optimizers such as Gurobi or CPLEX, which efficiently solve linear programming models. This approach can be found in several studies [50][56][61][49][46][54][62]. In these studies, exact historical price data is used in the optimization problem, meaning the results can be interpreted as a theoretical maximum revenue that could have been achieved if prices were fully known.

2.6.7 Dynamic Programming approaches

Another common approach to battery scheduling is to use Dynamic Programming (DP). DP approaches decomposes the problem into smaller decision-making problems and then solves and combine these. These often use the SoC and market prices as states and then find the best actions in terms of charge/discharge rate and bid sizes.

Several different DP approaches can be found in [63][64][65]. In [63], stochastic DP was evaluated and found that frequency regulation services were by far the most lucrative service and that model performance was not that sensitive to price forecast accuracy. In [64], the authors tried DP on several different timescales and then combined the different policies. The work found problems with scaling the resolution and the state space of the problem and, therefore, used approximation techniques such as singular value decomposition. In [65] an advanced variation of DP is used which does not have to discretize the state and the action space. The work focuses more on optimizations for PV and finds that the model performs very well, but only above a certain compute threshold and below this threshold the model performs poorly.

A common challenge in these DP works is that they have to deal with the curse of dimensionality that their state space quickly becomes very large and maybe even intractable to compute.

2.6.8 Price Forecasting

In [66], SARIMAX models were developed to forecast prices for the FCR-D Up, FCR-D Down, and FCR-N services. The models were based on historical market data and assumptions about future developments, derived from an interview with an industry expert and current market reports. The paper focuses on predicting overarching price trends between 2024 and 2030, and suggests that prices for the frequency regulation services will continue to decrease sharply. For FCR-N and FCR-D Up, the authors expect the prices to drop by 75-80%. An even bigger decrease is expected for FCR-D Down, where the authors estimate the prices to plummet over 95% during the same time period. The findings indicate that relying solely on frequency regulation services will become significantly less profitable in the coming years, which strengthens the need for batteries to find new revenue streams.

When it comes to optimizing BESS operation between different FCR markets and energy arbitrage, overarching price trends are not sufficient. More granular predictions on market prices for each hour are needed. This is attempted by [45], who trains and compares an Artificial Neural Network model, a LSTM model, against a naive baseline model that uses the prices of the previous day as forecasts. Interestingly, the findings suggest that the naive model performs best overall. Furthermore, the study suggests that marginal capacity compensation prices are harder to predict than average compensation prices (which were reported before the introduction of the marginal pricing dynamic in the FCR markets).

2.7 Related Work Summary

Previous research generally indicates that FCR services are suitable for BESS but that the price volatility of the Nordic markets is generally too low for energy arbitrage to be a competitive option. Peak shaving has also been identified as a promising strategy, as well as an increase in the value of BESS for grids that use power tariffs.

Different methods of varying complexity for modeling battery degradation are highlighted and it is found that when incorporating degradation into the optimization, degradation can be decreased significantly with quite minimal impact on revenue.

For multi-market optimization, methods of several degrees of complexity are analyzed. Firstly, rule-based algorithms for FCR, peak shaving, and PV self-consumption are covered. Secondly, more advanced models like LP and

DP are covered, these models have mostly assumed perfect price information, and it is identified that DP can struggle with large state spaces. Finally, price forecasting is highlighted. It is shown that simple baseline methods are hard to beat using models such as artificial neural networks and LSTMs.

The literature review highlights several research areas and several research gaps. Although there have been several studies on the use of BESS for FCR, energy arbitrage, and even a combination of these, no prior work has been identified that incorporates exact LER requirements, avoids the assumption of perfect price information, and accounts for power tariffs. This work is meant to fill this gap and provide a framework for developing realistic and deployable models for multi-market battery scheduling.

Chapter 3

Method

3.1 Data Collection

The frequency regulation market data used in this thesis are retrieved from Svenska Kraftnät's open data platform, Mimer, available at https://mimer.svk.se/PrimaryRegulation/PrimaryRegulationIndex. The data includes historical information on procured volumes and prices in the FCR markets (FCR-N, FCR-D Down, FCR-D Up). The prices for the FCR services are taken from the second auction (D-1), since, unlike the first auction, the second auction closes after the spot prices for the upcoming day are made available, which means that they can be used when deciding the BESS operation schedule.

The grid frequency data, which is used to calculate frequency regulation activations, is retrieved from Fingrid's open data platform, available at https://data.fingrid.fi/en/data?datasets=177. The data consists of historical frequency measurements taken at 3-minute intervals for the Nordic synchronous area. Since Sweden and Finland are part of the same synchronous electricity grid, they share a common frequency of the system.

The electricity price data are retrieved from Danish Energinet's data platform, Energi Data Service, available at https://www.energidataservice.dk/tso-electricity/elspotprices. The dataset contains historical day-ahead spot prices for different bidding zones, including the Swedish zones SE3 and SE4. In this thesis, the electricity prices from SE3 are used, as the supervising company Greenely has the highest volume of pre-qualified BESS assets in this zone. SE3 covers the middle parts of Sweden, including the Stockholm area.

Consumption data is retrieved from Greenely's internal systems. The data is filtered in order to only include customers living in villas and who have consumption data available for every hour of the studied time period. The final consumption data contains a total of 48 randomly selected households.

The time period of the data studied is chosen between 1 February 2024 and 31 December 2024. The reason January is excluded is because the marginal pricing dynamic of the frequency regulation markets was introduced on February 1 2024 [24].

Dataset	Source	Variables	Resolution	Data Points
FCR Market Prices	SvK (Mimer)	Capacity Compensation Price (€/MWh)	Hourly	3 × 8017
Grid Frequency	Fingrid	Frequency (Hz)	3-minute	183973
Spot Prices (SE3)	Energinet	Spot Price (€/MWh)	Hourly	8017
Consumption	Greenely	Consumption (kWh)	Hourly	48 × 8017

Table 3.1: Summary of datasets used in the thesis.

3.2 Assumptions

Price Taker. It is assumed that the aggregator behaves as a price taker in all electricity markets. This means that the bidding strategy of the aggregator does not influence the market clearing prices. This assumption is reasonable given the relatively small market share of an individual aggregator compared to the total market size. As a result, market prices are treated as exogenous inputs to the model.

Capacity Availability. As mentioned earlier, the FCR markets have a minimum bid capacity, and therefore the BESS must be aggregated together with other residential BESS using VPP technology to participate in frequency regulation. The thesis will continue with the assumption that sufficient residential BESS are available for aggregation in order to always meet the minimum bid size requirements. Since all batteries are assumed to be identical and sufficient units are available for aggregation, it is sufficient to model a single representative battery. The results can then be scaled to reflect the behavior of the entire VPP fleet.

Bid Acceptance. It is assumed that all submitted bids are accepted. This is a reasonable assumption as the marginal pricing dynamic of the frequency regulation markets creates the possibility of submitting bids with a price point close to zero, which are almost guaranteed to be accepted. Marginal pricing ensures that bids will be compensated with the same price as the most expensive accepted bid.

Aggregator Fee. A 50% aggregator fee is assumed to be deducted from all revenues earned through frequency regulation. The fee is based on the current fee at Supervising company Greenely. This creates a difference in perspective between optimizing for the individual BESS owner and for the aggregating company, as the aggregator will miss out on frequency regulation revenue when the BESS owner utilizes the BESS for purposes other than frequency regulation. In this work, all optimizations are performed in order to maximize the revenue of the BESS owner.

3.3 Optimization Model

In order to find the best battery policy, an optimization model was built.

Initially, both DP and MILP models were built to assess which approach was more suitable. One major difference between the two models was that in DP there is a need to program what the model can do. The MILP model instead looks at the constraints, and then the optimizer finds what it can do without breaking the constraints. When adding more markets, it was found that the DP approach was difficult to scale since it relied on finding all possible bid combinations for participation in all markets. The MILP model was instead easier to scale, as it only required adding a term to the utility function and program in the constraints of the new market. Therefore, only the MILP model was finished in the sense that it included all the markets studied.

The MILP model was built in Python using the Pyomo library and the Gurobi Optimizer. The implementation was aided by the AI-powered programming assistant GitHub Copilot.

3.3.1 Mathematical formulation

A. Definitions

Symbol	Definition
$t \in \mathcal{T}$	Index and set of timesteps in time period [$\triangle t$ = 1 hour]
P_{max}	Maximum power output [kW]
$S_{\text{min}}, S_{\text{max}}$	Min/max state of charge [kWh]
S_0	Initial state of charge at timestep 0 [kWh]
$\eta_{\text{cha}}, \eta_{\text{dis}}$	Charge/discharge efficiency
π_t^{spot}	Spot price at timestep t [€/kWh]
π_t^X	Capacity compensation price for FCR-X at timestep t, $X \in \{\text{N}, \text{DU}, \text{DD}\}$ [€/kW]
π^{tax}	Electricity tax [€/kWh]
C^{cyc}	Cycle cost per full cycle [€/cycle]
A_{fee}	Aggregator fee [%]
N^{cyc}	Total number of equivalent full cycles over the time period [cycles]

Table 3.2: List of symbols and their definitions

B. Decision Variables

S_t	State of charge at timestep t [kWh]
p_t^X	Power bid in FCR-X at timestep t [kW]
P_t^{cha}	Charging power at timestep t [kW]
P_t^{dis}	Discharging power at timestep t [kW]
P_t^{ref}	Reference power at timestep t [kW]

C. Objective Function

The objective is to maximize total revenue, which consists of revenue from price arbitrage, revenue from FCR capacity payments, and subtracting the cycle degradation costs. The objective function is defined as:

$$f = \max \quad R_{\text{arbitrage}}^{\text{tot}} + R_{\text{FCR}}^{\text{tot}} - C^{\text{tot}} \tag{3.1}$$

C.1. Revenue from Price Arbitrage

The revenue from price arbitrage comes from charging during low-price timesteps and discharging during high-price timesteps. This can be formulated as:

$$R_{\text{arbitrage}}^{\text{tot}} = \sum_{t \in \mathcal{T}} \left[\pi_t^{\text{spot}} \cdot P_t^{\text{dis}} - \left(\pi_t^{\text{spot}} + \pi^{\text{tax}} \right) \cdot P_t^{\text{cha}} \right] \tag{3.2}$$

Where π_t^{spot} is the spot price at timestep t [€/kWh]. P_t^{dis} is the discharging power at timestep t [kW]. P_t^{cha} is the charging power at timestep t [kW]. π^{tax} is the electricity tax [€/kWh].
The first term represents revenue from discharging at the spot price, while the second term subtracts the cost of charging (which includes the spot price and the electricity tax).

C.2. Revenue from FCR Capacity Payments

The revenue from FCR-D is based on the capacity reserved for frequency regulation services, not on the energy actually delivered. The revenue from FCR-N depends both on reserved capacity and the activated energy. The energy activation payment is estimated to be 0.036 of the capacity payment, which means that the total revenue from FCR-N is estimated as:

$$N^* = N \cdot 1.036 \tag{3.3}$$

The total FCR revenue is represented by:

$$R_{\text{FCR}}^{\text{tot}} = \sum_{t \in \mathcal{T}} \sum_{X \in \{\text{N}^*, \text{DU}, \text{DD}\}} A_{fee} \cdot \pi_t^X \cdot p_t^X \tag{3.4}$$

Where π_t^X is the capacity compensation price for FCR-X at timestep t [€/kW], with $X \in \{\text{N}^*, \text{DU}, \text{DD}\}$ corresponding to different types of frequency regulation. p_t^X is the power bid for FCR-X at timestep t [kW].

C.3. Degradation Cost

The degradation cost is incurred due to cycling of the battery, where each cycle incurs a cost. The total degradation cost is calculated as:

$$C^{\text{tot}} = C^{\text{cyc}} \cdot N^{\text{cyc}} \tag{3.5}$$

Where C^{cyc} is the cost per full cycle [€/cycle]. N^{cyc} is the total number of equivalent full cycles over the time period [cycles].
The number of cycles N^{cyc} can be estimated as:

$$N^{\text{cyc}} = N^{\text{cyc}}_{arb} + N^{\text{cyc}}_{FCR-N} \tag{3.6}$$

Where N^{cyc}_{arb} is the number of cycles from the arbitrage actions, i.e., charging and discharging, and N^{cyc}_{FCR-N} is the number of cycles from participating in FCR-N. N^{cyc}_{arb} is calculated as:

$$N^{\text{cyc}}_{arb} = \frac{1}{S_{\text{max}} - S_{\text{min}}} \sum_{t \in \mathcal{T}} \left(P^{\text{dis}}_t \cdot \eta_{\text{dis}} \right) \tag{3.7}$$

N^{cyc}_{FCR-N} is calculated as:

$$N^{\text{cyc}}_{FCR-N} = \frac{Bid_{FCR-N} \cdot 0.15}{(S_{\text{max}} - S_{\text{min}}) \cdot 2} \tag{3.8}$$

Where Bid_{FCR-N} is the total of all FCR-N bid sizes and the 0.15 factor is based on the average activation.

These equations linearly estimate the total number of equivalent cycles based on the energy throughput, where one full charge and discharge correspond to one cycle.
Note that these equations assume that FCR-D uses 0 cycles, this assumption is based on the fact that FCR-D has less than 0.1% average activation, because of how small the average activation is and the lack of an exact number, it is assumed to be 0 [24].

C.4. Final Objective Function

Combining all terms, the final objective function can be written as:

$$
\begin{aligned}
f = \max \quad & \underbrace{\sum_{t\in\mathcal{T}} \left[\pi_t^{\text{spot}} \cdot P_t^{\text{dis}} - \left(\pi_t^{\text{spot}} + \pi^{\text{tax}}\right) \cdot P_t^{\text{cha}}\right]}_{\text{Revenue from price arbitrage}} \\
& + \underbrace{\sum_{t\in\mathcal{T}} \sum_{X\in\{\text{N}^*,\text{DU},\text{DD}\}} \pi_t^X \cdot p_t^X}_{\text{Revenue from FCR capacity payments}} \\
& - \underbrace{C^{\text{cyc}} \cdot N^{\text{cyc}}}_{\text{Cycle degradation cost}}
\end{aligned}
\tag{3.9}
$$

D Battery Constraints

The operation of the battery is subject to several physical and technical constraints, including power limits, energy storage limits, and non-simultaneous charging and discharging. These are formulated as follows:

D.1. State of Charge (SoC) Dynamics

$$S_{\min} \leq S_t \leq S_{\max} \quad \forall t \in \mathcal{T} \tag{3.10}$$

The state of charge must always remain within the allowable energy storage range.

$$S_t = S_{t-1} + \eta_{\text{cha}} \cdot P^{\text{cha}}t - \frac{1}{\eta_{\text{dis}}} \cdot P_t^{\text{dis}} \quad \forall t \in \mathcal{T} \tag{3.11}$$

This equation updates the battery's SoC at each timestep, accounting for charging and discharging with respective efficiencies.

D.2. Power Limits

$$0 \leq P_t^{\text{cha}} \leq P_{\max} \qquad \forall t \in \mathcal{T} \tag{3.12}$$
$$0 \leq P_t^{\text{dis}} \leq P_{\max} \qquad \forall t \in \mathcal{T} \tag{3.13}$$

The battery's charging and discharging powers must be non-negative and cannot exceed the maximum inverter power.

D.3. Custom Power Limits

Custom power limits were created in order to evaluate the impact of limiting charging during both day and night hours. This was implemented using:

$$P_t^{\text{cha}} \leq \begin{cases} P_{\text{cha,night}}^{\max} & \text{if } t \bmod 24 \in \{22, 23, 0, 1, 2, 3, 4, 5\} \\ P_{\text{cha,day}}^{\max} & \text{if } t \bmod 24 \in \{6, 7, \dots, 21\} \end{cases}, \quad \forall t \in \mathcal{T} \tag{3.14}$$

More about custom power limits can be found in section 3.4.

D.4. No Simultaneous Charging and Discharging

To prevent the battery from charging and discharging at the same time, the following constraint is introduced:

$$P_t^{\text{cha}} \cdot P_t^{\text{dis}} = 0 \quad \forall t \in \mathcal{T} \tag{3.15}$$

This constraint ensures that the battery cannot discharge and charge simultaneously at any given timestep. Similar constraints can be added to prevent simultaneous FCR-D and FCR-N or FCR while charging or discharging in case the BESSs are not pre-qualified for such combinations.

E. FCR constraints for Limited Energy Resources

Since the BESS studied cannot sustain maximum power for more than 2 hours without depleting, they are considered a LER resource. This means SvK has additional constraints regarding how the resource can be used in FCR.

E.1 Reference power and frequency regulation:

The *reference power* P_t^{ref} defines the scheduled net power exchange of the battery for hour t, assuming no frequency regulation is activated. Any activation of frequency regulation, either FCR-N or FCR-D, will therefore be a *deviation* from this reference power. Reference power is defined as the difference between scheduled charging and discharging:

$$P_t^{\text{ref}} = P_t^{\text{cha}} - P_t^{\text{dis}}, \quad \forall t \in \mathcal{T} \tag{3.16}$$

A positive P_t^{ref} indicates net charging (import), while a negative value indicates net discharging (export). For example, if the battery is charging at full speed, an upward regulation signal (discharging) could first reduce charging to 0, then start discharging at full speed. The net effect on the grid would thus be higher than the maximum power capacity of the BESS, allowing it to place bids higher than its maximum power capacity during hours with a specified non-zero reference effect.

E.2. Capacity reservation rules:

LER units are also required to reserve additional capacity beyond the offered amount, to account for recovery needs and managing the state of charge during delivery hours:

- For FCR-D: 20% of the offered capacity must be reserved in the opposite direction.
- For FCR-N: 34% of the offered capacity must be reserved in both directions.

These requirements lead to the following power reservation constraints, where $p_t^{\text{N}}, p_t^{\text{DU}}, p_t^{\text{DD}}$ are the capacities offered for FCR-N, FCR-D Up, and FCR-D Down, respectively, and P_t^{ref} is the reference power the battery is scheduled to sustain for the whole hour. To ensure capacity availability after applying the reservation rules, the following must hold:

$$1.34\,p_t^{\text{N}} + p_t^{\text{DU}} + 0.2\,p_t^{\text{DD}} \leq P_{\text{max}} + P_t^{\text{ref}}, \qquad \forall t \in \mathcal{T} \tag{3.17}$$
$$1.34\,p_t^{\text{N}} + p_t^{\text{DD}} + 0.2\,p_t^{\text{DU}} \leq P_{\text{max}} - P_t^{\text{ref}}, \qquad \forall t \in \mathcal{T} \tag{3.18}$$

E.3. Endurance requirements:

To be qualified for participation, LER resources must be able to deliver:

- At least 60 minutes of continuous full activation in both up- and down-regulation for FCR-N.
- At least 20 minutes of continuous full activation for FCR-D Up and FCR-D Down.

Since batteries have limited energy, it is crucial to ensure that the state of charge (SOC) remains within limits even when the FCR bids are fully activated. The SoC is updated based on the reference power P_t^{ref} and must account for possible full FCR activations in both directions. The following constraints ensure that the battery has sufficient energy to handle the required activation duration in different scenarios:

If no frequency regulation is required the battery will charge/discharge following its reference power:

$$S_{\min} \leq S_{t-\Delta t} + P_t^{\text{ref}} \leq S_{\max}, \qquad \forall t \in \mathcal{T} \tag{3.19}$$

If the FCR-D Down bid is fully activated (meaning the battery has to charge electricity from the grid) for 20 minutes. Any activation of FCR-D Down implies that the FCR-N bid is fully activated in the downward (charging) direction. See Figure 2.2:

$$S_{\min} \leq S_{t-1} + \left(P_t^{\text{ref}} + p_t^{\text{N}} + p_t^{\text{DD}}\right) \cdot \tfrac{1}{3} \leq S_{\max}, \qquad \forall t \in \mathcal{T} \tag{3.20}$$

If the FCR-D Up bid is fully activated (meaning the battery has to release electricity back to the grid) for 20 minutes. An activation of FCR-D Up implies that the FCR-N bid is fully activated in the upward (discharging) direction:

$$S_{\min} \leq S_{t-1} + \left(P_t^{\text{ref}} - p_t^{\text{N}} - p_t^{\text{DU}}\right) \cdot \tfrac{1}{3} \leq S_{\max}, \qquad \forall t \in \mathcal{T} \tag{3.21}$$

If the FCR-D Down bid is fully activated for 20 minutes and the FCR-N bid is fully activated for the whole hour:

$$S_{\min} \leq S_{t-1} + P_t^{\text{ref}} + p_t^{\text{N}} + \tfrac{1}{3} \cdot p_t^{\text{DD}} \leq S_{\max}, \qquad \forall t \in \mathcal{T} \tag{3.22}$$

If the FCR-D Up bid is fully activated for 20 minutes and the FCR-N bid is fully activated for the whole hour:

$$S_{\min} \leq S_{t-1} + P_t^{\text{ref}} - p_t^{\text{N}} - \tfrac{1}{3} \cdot p_t^{\text{DU}} \leq S_{\max}, \qquad \forall t \in \mathcal{T} \tag{3.23}$$

Here, the factor $\frac{1}{3}$ corresponds to the 20-minute energy delivery requirement for FCR-D, compared to 60 minutes for FCR-N. These expressions ensure that the SoC remains within its limits regardless of the combination of reference power and regulation service activations.

3.4 Power Tariffs and Peak Shaving

The specifics regarding how the power tariff is calculated make it difficult to model and optimize for using MILP. On top of this, precise modeling in order to find optimal policies regarding FCR, energy arbitrage and peak shaving would require precise modeling and predictions of individual households consumption patterns. Due to all these difficulties, no advanced peak shaving strategies were directly programmed into the model directly. However, different heuristic-based strategies were tested to see how the battery policy would impact power tariffs and revenue.

A strategy $\mathbf{S_{x\text{-}y}}$ corresponds to setting:

$$P^{\max}_{\text{cha,day}} = \frac{x}{100} \cdot P_{\max} \tag{3.24}$$

$$P^{\max}_{\text{cha,night}} = \frac{y}{100} \cdot P_{\max} \tag{3.25}$$

The different strategies tested were as follows:

- $\mathbf{S_{100\text{-}100}}$ (no restrictions, this strategy is used if not indicated otherwise)
- $\mathbf{S_{50\text{-}100}}$ (limit to 50% charge speed daytime)
- $\mathbf{S_{0\text{-}100}}$ (prohibit charging daytime)
- $\mathbf{S_{0\text{-}50}}$ (prohibit charging daytime, limit to 50% charge speed nighttime)

In addition to these, the same strategies were also tested but with the added limitation of only participating in FCR-N during the night hours. For these strategies, the numbers in the subscript indicate the same charging limitations as described earlier and the * in the superscript indicates the FCR-N daytime limitation. These strategies are as follows:

- $\mathbf{S^{*}_{100\text{-}100}}$ (prohibit FCR-N during daytime)
- $\mathbf{S^{*}_{50\text{-}100}}$ (prohibit FCR-N during daytime, limit to 50% charge speed daytime)
- $\mathbf{S^{*}_{0\text{-}100}}$ (prohibit FCR-N during daytime, prohibit charging daytime)
- $\mathbf{S^{*}_{0\text{-}50}}$ (prohibit FCR-N during daytime, prohibit charging daytime, limit to 50% charge speed nighttime)

The strategies are then evaluated by looking at two metrics. Firstly, how much do these restrictions decrease the revenue generated compared to the baseline strategy $\mathbf{S_{100\text{-}100}}$ which does not limit the battery at all. Secondly, how much would the policy generated using these restrictions increase the power tariffs over the studied period. This is evaluated by looking at the consumption data of 48 Greenely customers (all living in villas since these are the main target group for residential batteries). The total amount of power tariffs they would have paid if power tariffs had already been implemented during the studied time is then calculated, as well as the total amount of power tariffs if the battery charging and discharging pattern is added to the household consumption. The difference between these are then calculated and divided by the number of households in order to get the increase of power tariff due to battery actions.

Battery consumption was calculated by looking at how much the battery was charging or discharging and by looking at how much FCR would have consumed during each hour. The FCR consumption was calculated by looking at historical frequency data and inferring the activations.

3.5 Price Forecasting

Price forecasting was implemented using five different methods, these include three different naïve baseline methods and two statistical models. All forecasting was done regarding the D-1 auction. The models are trained and evaluated using a time-series split, where the last three months of 2024 make up the test data.

3.5.1 Baseline - $\mathbf{B_{yest}^{avg}}$

A baseline was created by predicting the average of yesterday's prices for both FCR-N and FCR-D for all upcoming 24 hours. This means that for each individual market, all hours of the upcoming day will be set at the same price point as the average of yesterdays prices for that market. This baseline will be referred to as B_{yest}^{avg}.

3.5.2 Baseline - $\mathbf{B_{yest}}$

The second baseline predicts the price for each hour of the upcoming day using the price from the same hour on the previous day. Compared to B_{yest}^{avg}, which uses a single average value from the previous day for all 24 hours, B_{yest} produces hourly varying predictions. This baseline will be referred to as B_{yest}.

3.5.3 Baseline - $\mathbf{B_{D-2}}$

The third baseline model is based on the prices of the first FCR auction, which closes at 00:30 the day before planned delivery, five and a half hours before the day-ahead spot price market closes. This means that before the second FCR auction closes at 18:00 on the same day, the final prices of the first auction are already available. In this baseline model, the predicted prices for each hour of the upcoming day are the same as the final prices for each hour of the first auction. This baseline will be referred to as B_{D-2} where D-2 refers to the auction D-2 which is the first auction.

3.5.4 ARIMA

ARIMA was implemented with the objective of predicting future prices. The model was tasked with predicting 24 hours in advance. For each market, a hyperparameter grid-search was conducted in order to find the most suitable values of p, d and q for each market. All combinations of p, d, and q where $p, d, q \in \{0, 1, 2, 3, 4, 5\}$ were tested. The hyperparameters which led to the model with the lowest Mean Absolute Error (MAE) were selected as most suitable for predicting prices in that market.

3.5.5 Prophet

Prophet was implemented to predict hourly prices 24 hours ahead for the different FCR markets. A separate model was trained for each market. The default configuration of Prophet was used.

To improve accuracy, the prices from the early auction (D-2) were added as regressors. These prices are known before the second auction closes and can therefore serve as useful input when predicting the prices of the second auction.

In some cases, Prophet produced negative price predictions. All negative predictions were set to zero, as negative prices are not valid in the FCR capacity compensation markets. To simulate a real-world scenario, the model was re-trained each day using all available historical data up to the forecast date.

3.5.6 Evaluation

To assess the performance of the models, four error metrics were used: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), MAE and Mean Absolute Percentage Error (MAPE). They are defined as follows:

- **Mean Squared Error (MSE)** is defined as the average of the squared differences between the predicted FCR prices $\hat{y}_t$ and the actual FCR prices y_t:

$$\text{MSE} = \frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2$$

- **Root Mean Squared Error (RMSE)** is the square root of the MSE and provides an error value in the same unit as the FCR prices:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

- **Mean Absolute Error (MAE)** is the mean of the absolute differences between the predicted and actual FCR prices:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

- **Mean Absolute Percentage Error (MAPE)** expresses the average absolute forecast error as a percentage of the actual FCR prices:

$$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|$$

Here, n denotes the number of forecasted hours, y_t the actual FCR market price at hour t, and $\hat{y}_t$ the corresponding predicted price.

3.6 Parameters

The parameters are chosen to reflect a typical residential BESS for a Greenely customer. The BESS has a C-rating of 1, which means that it can fully charge or discharge in one hour. The complete set of parameters used can be found in Table 3.3.

Table 3.3: Simulation Parameters

Parameter	Value
Energy Storage Capacity	10 kWh
Max Output Power	10 kW
Min/Max SoC (%)	10% - 90%
Charging Efficiency	90 %
Discharging Efficiency	90 %
Electricity Tax	€0.0428/kWh [36]
Power Tariff	€8.125/kWh
Average FCR-N Activation	15 %
Average FCR-N Activation Remuneration (of FCR-N capacity revenue)	3.6 % [27]
Battery purchasing cost	€15 000
End-Of-Life (EOL) Battery Cycles	10 000 Cycles
EOL Battery Calendar life	10 Years
Cycle Cost	€1.5/Cycle
Aggregator fee	50 %
Date range	1 Feb 2024 - 31 Dec 2024

Chapter 4

Results and Analysis

In this chapter, the results and performance of the implemented optimization model and the price prediction models are presented. Examples of the optimal battery schedule for a specific day are presented to showcase the various possible bid combinations and typical arbitrage opportunities. Additionally, the performance of the optimization model is evaluated when combined with the sub-obtimal price predictions. Lastly, the BESS operations effect on power tariffs under different custom charging-discharging and FCR constraints is presented.

4.1 Multi-market optimization

4.1.1 Strategy Examples

To evaluate the performance of different strategies for battery operation in electricity markets, single-market strategies are compared with a combined multi-market optimization. The analyzed strategies are:

- **Arbitrage only:** The battery only engages in price arbitrage, charging when prices are low and discharging when prices are high. Example battery usage based on spot prices is displayed in Figure 4.1.

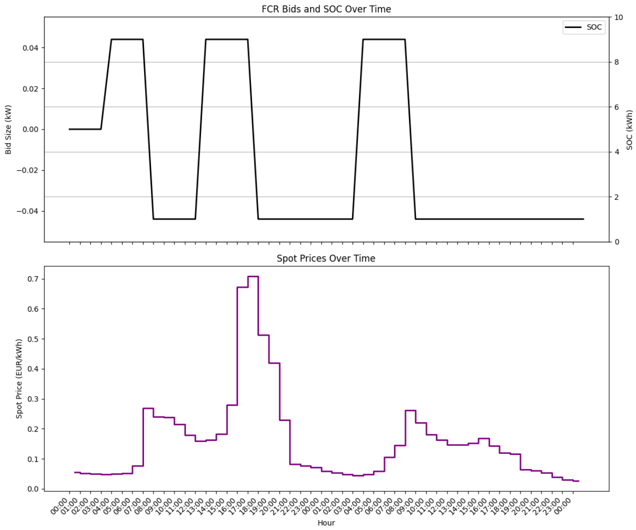


Figure 4.1: Arbitrage-only strategy on December 12 & 13 2024.

- **FCR-D Up & Down only:** The battery SoC is kept close to 50%, providing FCR-D Up and FCR-D Down simultaneously. This was the bidding strategy used by supervising company Greenely when this work started. An example is displayed in Figure 4.2.

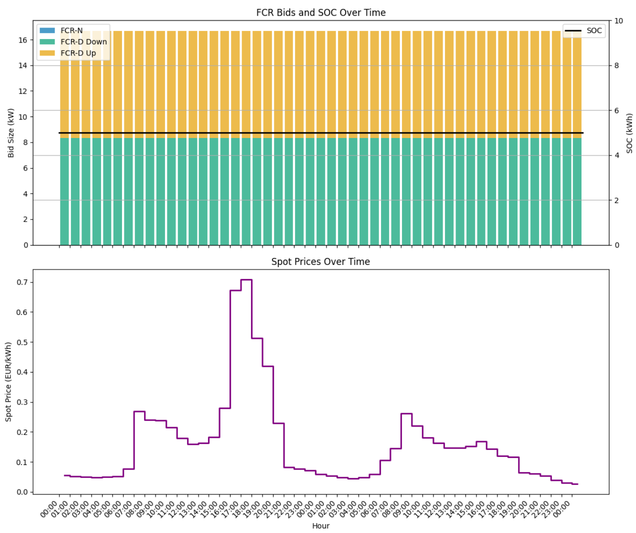


Figure 4.2: Simultaneous FCR-D Up and Down strategy on December 12 & 13 2024.

- **FCR-N only:** The battery only participates in FCR-N. An example is displayed in Figure 4.3

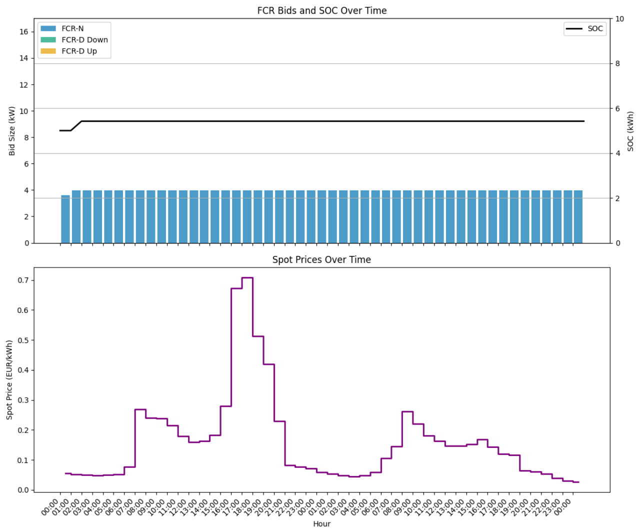


Figure 4.3: FCR-N-only strategy on December 12 & 13 2024.

- **Multi-optimization:** The battery operation is optimized across all services, balancing between arbitrage, FCR-N, FCR-D Up, and FCR-D Down, maximizing economic return. An example of battery usage and bidding based on spot price and FCR prices is displayed in Figure 4.4.

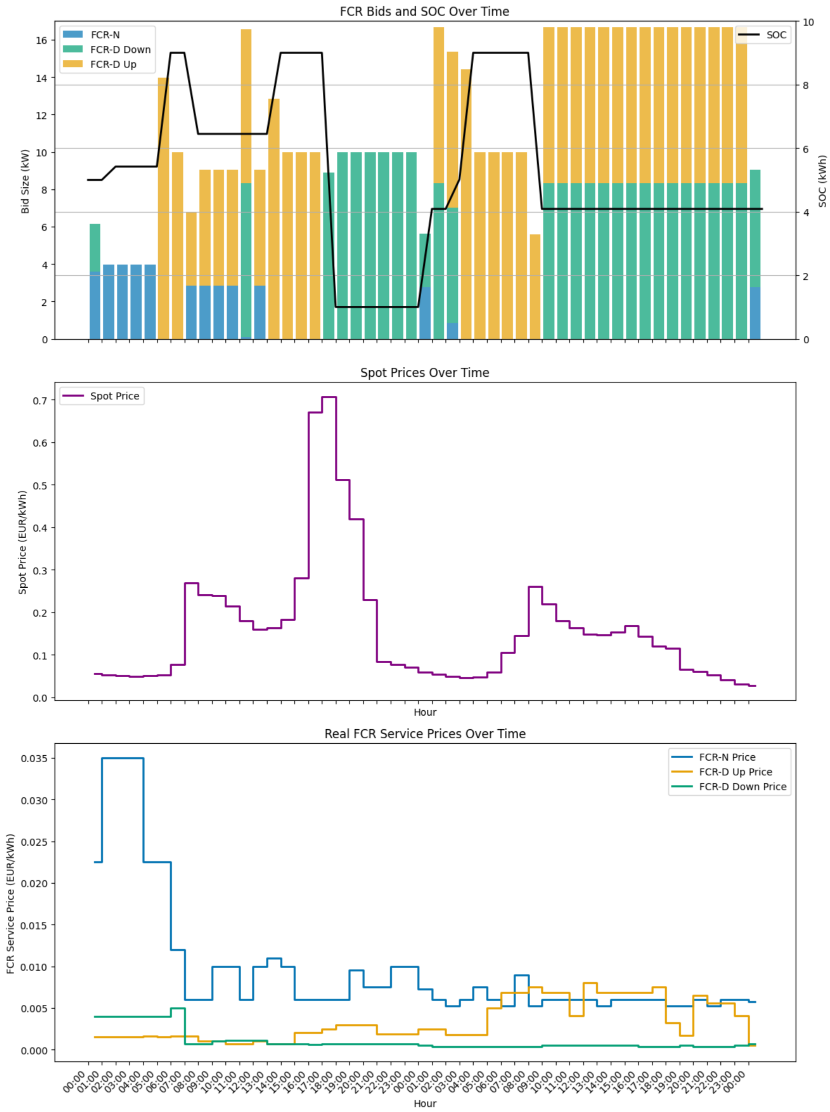


Figure 4.4: Battery operation under multi-market optimization combining arbitrage and FCR services. Dec 12 & 13, 2024

4.1.2 Revenue under perfect price information

Table 4.1 presents a comparison of the different bidding strategies under perfect foresight of market prices. For each strategy, it reports the total revenue with and without degradation costs. The degradation cost is always included in the objective function during optimization, however, for the "without degradation" column, the total number of equivalent full cycles times the cycle cost is added to the optimized revenue. The table also shows the number of equivalent full cycles and the relative change in aggregator revenue compared to the baseline strategy (FCR-D Up Down every hour).

Table 4.1: Revenue and operational comparison across strategies (perfect foresight, €1.5/Cycle degradation costs).

Strategy	Revenue w/ Deg (€)		Revenue w/o Deg (€)		Battery Cycles	Aggregator Revenue (%)
FCR-D Up & Down (All hours)	1 659.6	(0.0%)	1 659.6	(0.0%)	N / A	±0%
Arbitrage only	5.2	(-99.7%)	11.3	(-99.3%)	4.1	−100.0%
FCR-N Only (All Hours)	370.8	(-77.7%)	819.1	(-50.6%)	298.9	−50.6%
Multi-market optimization	**2 057.7**	**(+24.0%)**	**2 138.9**	**(+28.9%)**	54.2	+28.9%

The multi-market optimization increases BESS revenue by 24% when accounting for degradation costs. If degradation is added back after the optimization, revenue increases by 28.9%, both for the BESS owner and the Aggregator. Looking at the table, none of the strategies surpasses 1000 cycles per year, meaning that the BESS will reach its EOL-due to calendar aging before reaching EOL due to the number of cycles.

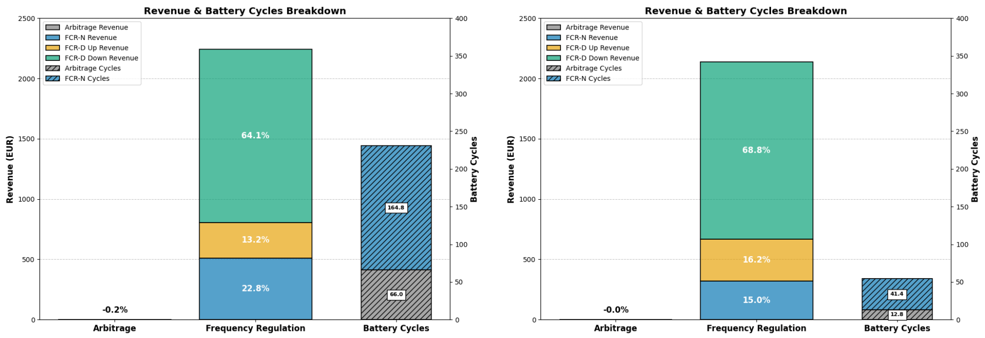
Table 4.2: Revenue and operational comparison across strategies (perfect foresight, €0/cycle degradation costs).

Strategy	Revenue w/ Deg (€)		Revenue w/o Deg (€)		Battery Cycles	Aggregator Revenue (%)
FCR-D Up & Down (All hours)	1 659.6	(0.0%)	1 659.6	(0.0%)	N/A	±0%
Arbitrage only	40.1	(-97.6%)	40.1	(-97.6%)	77.4	−100.0%
FCR-N Only (All Hours)	819.1	(-50.6%)	819.1	(-50.6%)	298.9	−50.6%
Multi-market optimization	**2 236.9**	**(+34.8%)**	**2 236.9**	**(+34.8%)**	230.9	+35.0%

In the case of a cycle cost of zero, revenue using the multi-market optimization scheme increases by 34.8% compared to the baseline bidding strategy of equal FCR-D Up & Down bids every hour.

4.1.3 Revenue Breakdown

To illustrate the different revenue sources when using multi-market & arbitrage optimization, Figure 4.5 compares the case with degradation costs of (€0/cycle) to the case with degradation costs of €1.5/cycle.



(a) €0/cycle degradation cost

(b) €1.5/cycle degradation cost

Figure 4.5: Revenue sources using multi-market optimization (perfect foresight) under different degradation cost assumptions.

The graphs illustrate that the revenue from price arbitrage accounts for less than 0.1 % of the total revenue. This suggests that frequency regulation is still the most profitable revenue source for a residential BESS, with the majority of the frequency regulation revenue coming from FCR-D Down.

By introducing a cycle cost of €1.5/Cycle in the objective function, battery cycling can be reduced by roughly 77 %, with only a 4.4 % reduction in total revenue. In both cases, the total number of cycles is less than 1000 cycles / year, meaning that the battery will reach its rated calendar EOL before reaching its battery cycles EOL. Applying a zero-cost assumption for cycles might therefore be reasonable in practice. The following graphs will assume a cycle cost of zero.

4.1.4 Bid distributions

To understand the bidding behavior across different markets, the distribution of bid sizes submitted by the BESS are analyzed. Figure 4.6 shows a histogram of bid sizes across the considered markets, illustrating the frequency of different bid volumes.

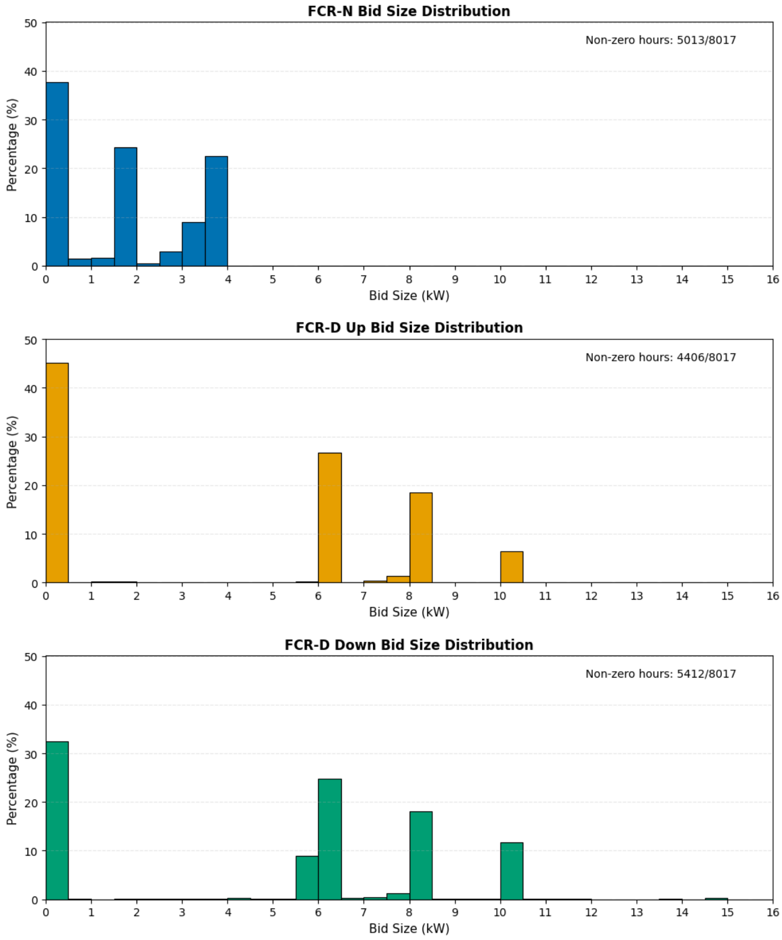


Figure 4.6: Histogram of bid sizes across different FCR markets using the multi-market optimization strategy.

The bid size distributions illustrate a few common bidding options. For FCR-N, the most common non-zero bid option was a bid between 1.5 and 2 kW, as this allows simultaneous bidding in the FCR-D markets, which can be seen in Figure 4.4. When bidding only in FCR-N, the maximum bid size is 4 kW for the modeled BESS, which can be seen in the second non-zero peak of the FCR-N histogram. For FCR-D, a common option is to bid 6 kW. This allows simultaneous bidding in FCR-N. Bidding between 8 and 8.5 kW is another common option, which allows for equal bids in both FCR-D Up & Down simultaneously. A bid size of 10 kW in FCR-D can be achieved when participating in only one of the markets, which is another alternative. Lastly, a few cases where the BESS places bids higher than the BESS power capacity can be noted. This is possible when changing the reference effect during certain hours. Examples of the various bid combinations can be seen in Figure 4.4.

Figure 4.7 presents a pie chart summarizing the relative share of varying combinations of multi-market participation during the same delivery hour.

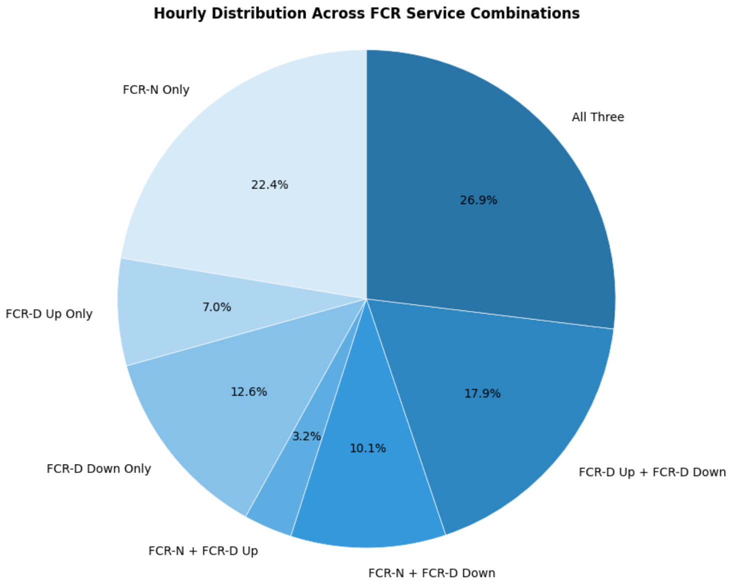


Figure 4.7: Relative share of different bid combinations

The most common bid option in the optimal strategy is to participate in all three FCR markets at the same time. Two other common options are to participate only in FCR-N, as well as doing FCR-D Up & Down simultaneously. FCR-N is also sometimes combined with FCR-D Down or FCR-D Up.

4.2 Price forecasting

Table 4.3: Optimal ARIMA hyperparameters found via grid search

Series	**p**	**d**	**q**
FCR-D Up	2	1	2
FCR-D Down	5	1	4
FCR-N	2	1	5

Table 4.4: Performance comparison of ARIMA, Prophet, and baseline models for each FCR service

(a) FCR-D Up

Metric	ARIMA	Prophet	B_{yest}^{avg}	B_{yest}	B_{D-2}
MSE	695.0	689.8	884.7	1188.6	**686.4**
RMSE	26.4	26.3	29.7	34.5	**26.2**
MAE	7.3	8.4	7.9	8.0	**6.2**
MAPE	312.2%	395.6%	309.0%	**197.2%**	211.2%

(b) FCR-D Down

Metric	ARIMA	Prophet	B_{yest}^{avg}	B_{yest}	B_{D-2}
MSE	81.4	574.6	110.4	141.0	**81.1**
RMSE	**9.0**	24.0	10.5	11.9	**9.0**
MAE	**3.3**	13.4	3.6	3.8	**3.3**
MAPE	163.1%	870.6%	135.5%	**131.4%**	153.9%

(c) FCR-N

Metric	ARIMA	Prophet	B_{yest}^{avg}	B_{yest}	B_{D-2}
MSE	199.2	288.1	217.3	341.2	**151.6**
RMSE	14.1	17.0	14.7	18.5	**12.3**
MAE	9.4	12.4	8.9	10.5	**7.1**
MAPE	49.3%	64.6%	41.3%	48.2%	**33.4%**

In general, it seems like B_{D-2} (prices from the early auction) works the best, having the lowest error in three out of four metrics for FCR-D services and all four metrics for FCR-N, as presented in Table 4.4. B_{yest} has the best MAPE for the FCR-D services while ARIMA is tied for the best MAE and RMSE (note, however, that the tie in RMSE is just due to rounding, if more decimal places would have been used, B_{D-2} is better which can be seen since it has a lower MSE).

Another thing worth noting is that the errors are generally the highest for FCR-D Up followed by FCR-N and then FCR-D Down. FCR-N does however have a significantly lower MAPE compared to the FCR-D services. The Prophet model had the worst overall performance, with exceptionally high error rates for the FCR-D Down market.

4.3 Combination of mutli-market optimization and price forecasting

4.3.1 Revenue using predicted prices

In order to assess model performance in a real world setting, the model was used to plan the battery schedule using the predicted FCR capacity market prices. The evaluation was done by creating the schedule based on predicted prices, but evaluated using the actual historical prices. The real spot prices are available to the model as they are released before the D-1 auction and thus are available in a real world setting.

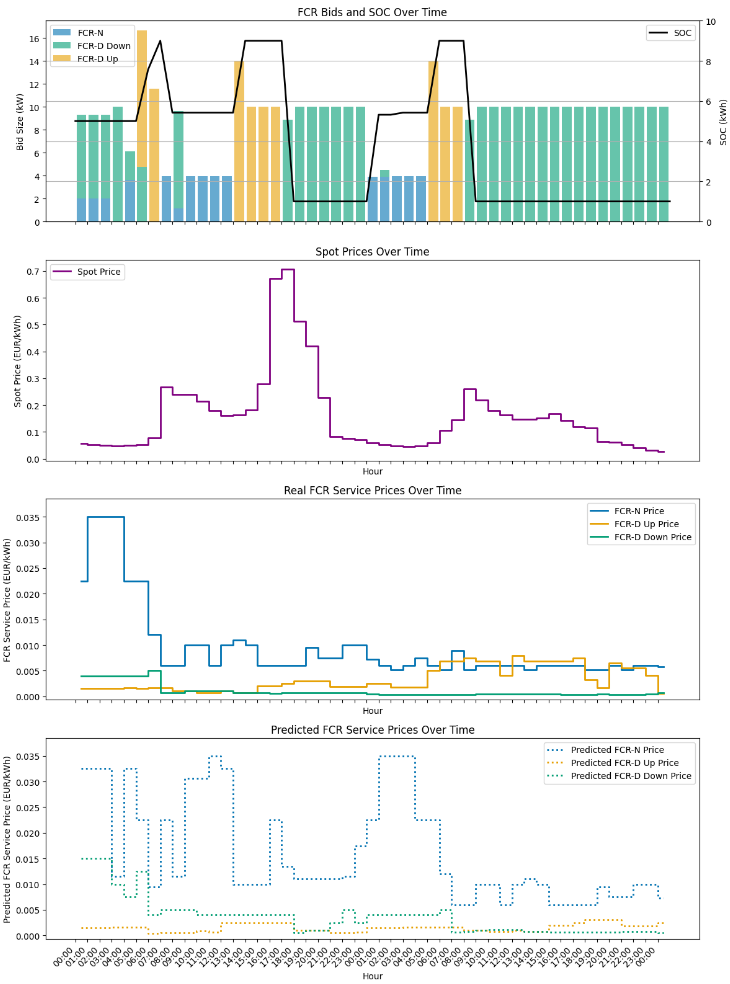


Figure 4.8: Example of battery schedule optimization using predicted FCR prices December 12 & 13, 2024. B_{yest} is used.

Figure 4.8 showcases the battery operation when scheduled using hourly capacity compensation prices from the day before (B_{yest}). The operation can be compared with the optimal operation (using perfect price knowledge) shown in Figure 4.4.

Table 4.5: Revenue and operational comparison across strategies. €0/Cycle degradation costs

Strategy	Revenue (€)		Battery Cycles	Aggregator Revenue (%)
Basic Strategies (evaluated using historical market prices)				
FCR-D Equal Up & Down (All Hours)	1 659.6	(0.0%)	N/A	$\pm 0\%$
Arbitrage only	40.1	(-97.6%)	77.4	-100.0%
FCR-N Only (All Hours)	819.1	(-50.6%)	298.9	-50.6%
Multi-Market Optimization (scheduled on predicted prices & evaluated on historical market prices)				
Price Prediction: Perfect Foresight	2 236.9	(+34.8%)	230.9	+35.0%
Price Prediction: B_{yest}^{avg}	1 819.0	(+9.6%)	199.3	+8.6%
Price Prediction: B_{yest}	1 720.9	(+3.7%)	227.8	+3.8%
Price Prediction: B_{D-2}	1 824.1	(+9.9%)	201.9	+8.9%
Price Prediction: Prophet	1 772.1	(+7.4%)	168.6	+6.9%
Price Prediction: ARIMA	**1 858.2**	**(+12.0%)**	196.0	+10.9%

The results using predicted market prices shown in Table 4.5 show that even without perfect foresight of frequency regulation market prices, the multi-market optimization strategy can still beat the current bidding strategy of equal FCR-D Up & Down bids every hour. Compared to having perfect foresight, the revenue is lower, which is expected. Out of the different predictions, ARIMA generates the highest total revenue. This is quite surprising since B_{D-2} generally had the best prediction results. It can also be seen that B_{yest} seems to work quite poorly compared to other methods, generating approximately three to seven percentage points less revenue for both the battery owner and the aggregator compared to the other prediction methods.

4.4 Revenue and Power Tariffs

To evaluate the value of batteries in more realistic scenarios, one also has to consider the power tariffs. This section evaluates how the strategies $\mathbf{S_{100\text{-}100}}$, $\mathbf{S_{50\text{-}100}}$, $\mathbf{S_{0\text{-}100}}$, $\mathbf{S_{0\text{-}50}}$, $\mathbf{S^*_{100\text{-}100}}$, $\mathbf{S^*_{50\text{-}100}}$, $\mathbf{S^*_{0\text{-}100}}$ and $\mathbf{S^*_{0\text{-}50}}$ impact the revenue generated as well as the power tariff.

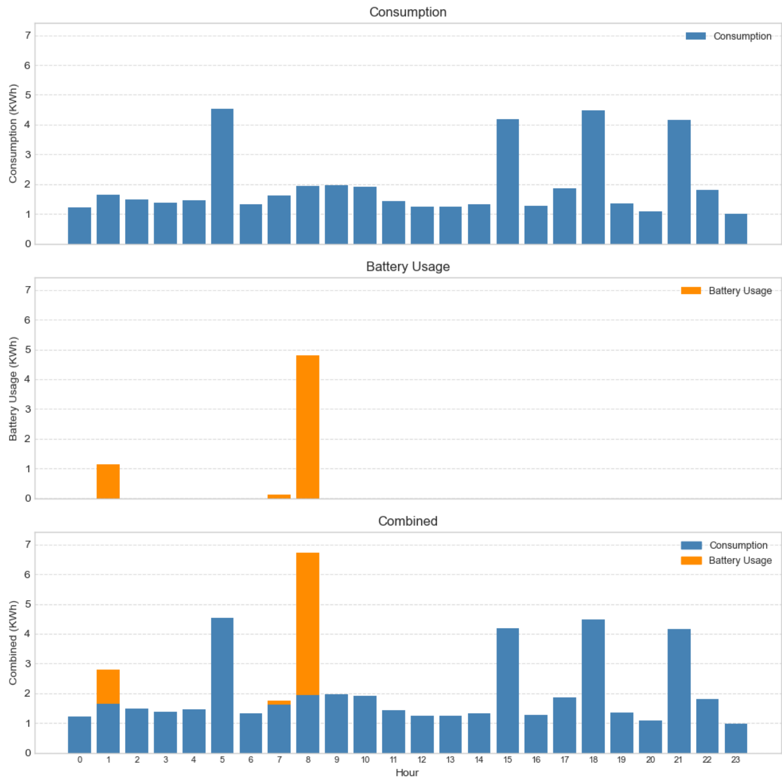


Figure 4.9: Example of battery usage creating a new power peak.

One risk of using batteries excessively is that it generates new power peaks which might lead to increased power tariffs. An example of a battery-generated peak is shown in Figure 4.9 where a peak more than 2 kWh higher than the previous daily peak is created. On the other hand, not allowing the battery to charge to decrease the risk of creating new power peaks could instead mean the battery risks missing out on market opportunities and decreasing the revenue generated.

Strategy	Revenue (€)	PT Increase (€, 95% CI)	PT Increase (%)	Net (Revenue - PT)
$S_{100\text{-}100}$	2 236.9 (±0.0%)	149.3 ± 19.1	+30.0%	2 087.6
$S_{50\text{-}100}$	2 236.6 (−0.0%)	142.9 ± 18.4	+28.7%	2 093.7
$S_{0\text{-}100}$	2 215.7 (−0.9%)	49.6 ± 6.6	+10.0%	2 166.1
$S_{0\text{-}50}$	2 215.6 (−1.0%)	48.5 ± 6.2	+9.7%	**2 167.1**
Limiting FCR-N participation to night hours only				
$S^*_{100-100}$	2 049.6 (−8.4%)	124.4 ± 18.3	+25.0%	1925.2
S^*_{50-100}	2 049.3 (−8.4%)	106.5 ± 17.0	+21.4%	1942.8
S^*_{0-100}	2 031.1 (−9.2%)	13.4 ± 6.0	+2.7%	2017.7
S^*_{0-50}	2 031.0 (−9.2%)	12.1 ± 5.6	+2.4%	2018.9

Table 4.6: Evaluation of different charging limitation strategies' impact on revenue and power tariffs. Using multi-market optimization with perfect foresight and €0/cycle degradation costs.

Strategy $S_{0\text{-}50}$ (which only allows charging during night hours and with a maximum 50% charging speed) has the highest net revenue when accounting for increased power tariffs. The total revenue using $S_{0\text{-}50}$ is decreased by 1.0% or €21.3 compared to the strategy with no charging limitations, while increasing the power tariff with only €49 compared to €149 for the no-limitations strategy. $S_{0\text{-}100}$ is a close second with just €1 less net revenue and with a very similar impact on both revenue and the increase in power tariffs compared to $S_{0\text{-}50}$.

Limiting FCR-N participation to night hours reduces the average power tariff by 25-50 € during the study period. However, this reduces total BESS revenue by around €185. This suggests that the overall loss of revenue from restricted participation in the FCR-N market outweighs the gains of lower tariffs.

Chapter 5

Discussion

5.1 Arbitrage Revenue

The results show that the revenue from electricity price arbitrage is negligible compared to the revenue generated from frequency regulation services. Even with a cycle cost of zero, arbitrage accounts for less than 0.1 % of the total revenue over the study period. The results align with the findings of similar studies, which suggest that electricity price fluctuations in Sweden are not high enough to generate any significant revenue from engaging in arbitrage [51][52]. The electricity tax is another factor that impacts the profitability of arbitrage, as there is a difference between the price paid for buying electricity and the price received when selling electricity back to the grid.

In the case with a cycle cost of zero, the total arbitrage revenue during the study period is negative, meaning that the BESS is effectively losing money from the buying and selling of electricity. The reason for this is likely due to the fact that the BESS adjusts its SoC in order to place optimal bids in the FCR markets, and might thus not necessarily charge or discharge solely based on the current spot prices. There are examples of the BESS charging and discharging simply in order to place higher bids in FCR-D markets by using a different reference effect. This phenomenon is explained in more detail in the next section.

5.2 Frequency Regulation Capacity Compensation Price Peaks

When it comes to FCR market prices, there are a couple of occurrences of extreme price peaks, where the price is significantly higher than the average price for that week or month. An example is on November 18, at 17:00, where the FCR-D Up regulation price peaked at 184 EUR/MW, compared to the monthly average of 2.7 EUR/MW [20]. The exact reasons behind this are not fully known but are likely related to the introduction of the marginal pricing dynamic early 2024. If there is a significant shortage of actors who submit bids for frequency regulation, TSO Svenska Kraftnät is forced to accept bids with very high price points in order to fill the procured volume. The marginal pricing dynamic then causes all bids to be compensated with the highest accepted price.

Single peaks like this can often make up a significant portion of the monthly FCR revenue. With the ability to vary the reference effect of the BESS, the BESS can utilize this to place bids higher than its maximum power capacity during these peak hours. In the optimal multi-market optimization strategy used in the thesis, the battery was scheduled to charge during the peak hour of November 18, at 17:00. This allowed the battery to place a bid of 14.4 kW in FCR-D Up instead of the usual maximum of 9-10 kW, effectively catching up to 44% more of the extreme price peak. Similar occurrences can be found in other months.

Unfortunately, extreme price peaks like these are very difficult to predict in advance. This might be one of the reasons why the optimal strategy using perfect price forecasts outperforms the strategies based on price predictions to such a large degree. It can also explain why using the hourly values of yesterday performs worse than using the average value of yesterday as predictions, since the BESS will act on any peaks from the day before, even though they probably will not happen again the next day. An example of this can be found in Figure 4.8.

5.3 Optimizing for the aggregator or individual BESS owner

The aggregator responsible for managing the VPP and submitting frequency regulation bids plays a crucial role for residential BESS owners. Without an aggregator and VPP technology, residential BESS would be too small to place bids in the frequency regulation markets. The aggregator can charge a fee for this service, in this study the fee is that they split the revenue from frequency regulation evenly between the asset owner and the aggregator. Thus, aggregators only have one source of income from the residential BESS (frequency regulation), while residential BESS owners can generate additional revenue from electricity price arbitrage. Whenever the BESS engages in arbitrage instead of frequency regulation, the aggregator misses out on FCR revenue. BESS owners also bear the costs of battery degradation and power tariffs, which aggregators do not.

As a consequence, maximizing the BESS revenue for the customer will not be the same as maximizing the revenue for the aggregator. In the thesis, revenue is maximized from an individual BESS owners point of view. Fortunately for the aggregator, arbitrage turns out not to be as profitable as frequency regulation during the study period, which means that the BESS owners multi-market optimization strategy including arbitrage and frequency regulation largely focuses on maximizing frequency regulation revenue. This explains why the increase/decrease in aggregator revenue is very close to the difference in customer revenue in Tables 4.1, 4.2, and 4.5.

However, if frequency regulation service prices continue to decrease, as they have been in the last months of 2024, arbitrage will begin to make up a larger portion of the total revenue for the BESS owner, which means the aggregator will miss out on more frequency regulation revenue. Implementing the multi-market optimization strategy proposed in the thesis might therefore very well lead to decreasing frequency regulation revenue for the aggregator, which they should be aware of. However, the strategy could easily be adapted to prioritize aggregator/FCR revenue by removing arbitrage profit and degradation costs from the objective function.

5.4 Forecasting

The performance of the price predictions varies for each frequency regulation market. The results suggest that FCR-N prices are the easiest to predict, although performance is relatively poor. The MAPE for the FCR-N predictions range between 30% and 65% as shown in Table 4.4. For FCR-D Up, MAPE values are extremely high, ranging between 200% and 400%, indicating the difficulty of accurately forecasting this volatile market. FCR-D Down forecasts also show poor accuracy, with MAPE values in the range of 130 % to 160 % with the exception that Prophet performs significantly worse with a MAPE of 870.6%.

Prophet produced the worst overall predictions, especially for the FCR-D Down market. One of the reasons might be that it captures the sharp decrease in market prices late 2024 as a continuous downward trend, and expects this trend to continue to price points below zero. The sudden drop in prices might explain why predicting future prices is challenging when using historical prices as training data. As shown in [45], which compared price prediction accuracy before and after the introduction of the marginal pricing dynamic, predicting marginal prices is harder. The introduction of marginal pricing might thus also contribute to the difficulty of accurately predicting market prices, as this leads to increased price volatility.

B_{D-2}, which uses the prices from the early auction, shows the best overall performance metrics. This finding aligns with the findings of other similar studies that attempt to predict FCR prices, where simple baselines seem to outperform or at least show comparable performance to more sophisticated prediction models.

Interestingly though, when the predictions are used as input for battery scheduling, the ARIMA predictions generate the highest total revenue. As shown in Figure 4.5, FCR-D Down accounts for the majority of the total revenue, and ARIMA has comparable performance as B_{D-2} in this market. However, no conclusions are drawn regarding why ARIMA leads to higher revenue despite generally having worse predictions, to explain this further investigation is needed.

The ARIMA-models hyperparameter search displayed in Table 4.3 shows that for all markets a d value of 1 was found to be the best. This is consistent with the observation that FCR prices have steadily decreased throughout the year (a d value of 1 means that the trend is removed through one step of differencing). Another observation is that none of the hyperparameters found is 0, which means all ARIMA parts (AM, I and MA) are used. This suggests that for the price prediction task in all markets ARIMA outperforms some simpler models (for example ARIMA with $d = 0$ is just a simpler model called ARMA and since 0 was not optimal in the hyperparameter search ARIMA seems more suitable than ARMA). The natural follow-up question is, however, is ARIMA complex enough or would even more complex models work better? The high p and q values found in the search for some markets along with large errors suggest that the underlying time series is complex and therefore it may be valuable to look into more complex models.

5.5 Multi-market optimization

The multi-market optimization highlights the complexity of optimal battery scheduling and frequency regulation market participation. As can be seen in Figures 4.7 and 4.4, the optimal strategy often includes bids in multiple frequency regulation markets at the same time. As other studies have shown [50], a common strategy is to have the SoC close to 50 % and place bids in both FCR-D Up and FCR-D Down simultaneously. The findings in this thesis show that an even more common option (26.9 % of all hours) is to participate in all three FCR markets (FCR-N, FCR-D Up & FCR-D Down) at the same time.

One reason behind this might be that the FCR-D prices have dropped significantly in the second half of 2024, rendering the FCR-N market more profitable in comparison. The difference in LER requirements for FCR-N and FCR-D also creates an interesting dynamic, where for FCR-N, the limiting factor restricting the maximum bid size is the storage capacity of the BESS. For FCR-D however, the limiting factor is the maximum power output of the BESS. This is caused by the varying endurance requirements for the two different markets, where FCR-N requires being able to sustain 60 minutes of full activation and FCR-D only requires being able to sustain 20 minutes. By placing smaller bids in all three markets, the BESS can instead be limited by both the storage capacity and the power output, meaning the BESS is utilized at the very edge of its capabilities.

An interesting observation is that when the model is run with price predictions instead of perfect price information, the number of cycles used drop by about 12 to 27 percent, with the exception of B_{yest} (yesterdays prices). This can be seen in Table 4.5. One reason for this could be that the other methods are in some sense smoothing/averaging previous prices (with the exception of using prices from D-2, however, these prices there are generally more stable). This means that the perfect prices and B_{yest} have significantly more peaks in the prices, for example, in Figure 4.8 one can see that B_{yest} predicts peaks in FCR-N prices, peaks that are not realized. Since the spot prices are the same for the different models, the increased number of cycles instead stems from the optimizer trying to change the reference effect when price peaks occur/are predicted, thus creating more cycles.

5.6 Power Tariffs and Revenue

In this work, all strategies tested increased the average power tariff, as shown in Table 4.6. Interestingly, both $S_{0\text{-}100}$ and $S_{0\text{-}50}$ significantly decreased the power tariff (compared to $S_{50\text{-}100}$ and $S_{100\text{-}100}$) while having a very small impact on the revenue which lead to the second highest and highest net earnings respectively. It may seem unintuitive that completely prohibiting charging of the battery for the majority of the day has such a small impact on revenue. However, by looking at Figure 4.5, this may make more sense. The data shows that arbitrage makes up a practically negligible part of the total revenue. Arbitrage does, however, make up a significant part of the battery cycles which can be seen in the left plot of Figure 4.5. Because the power tariffs only base the entire tariff on the highest consumption hours, it means that even if one does arbitrage only a few times a month, they will still likely increase the power tariff significantly. All this while earning a tiny share of total revenue. Therefore, by setting limits to the charging, only a few actions which have little impact on revenue but large impact on the power tariffs are removed. This explains why setting hard limits on how the battery is allowed to charge during the day may be a good strategy.

Contrary to this, Table 4.6 shows that limiting FCR-N quite consistently decrease the generated revenue with about €185 or about 8.3 percentage points compared to the corresponding non FCR-N limiting strategies. This decrease in revenue is significantly larger than the revenue loss when limiting the charging, this can be explained by looking at the revenue breakdown in Figure 4.5, where FCR-N contributes 22.8% of the total revenue in the €0 cycle cost case.

In the same figure, it can also be seen that FCR-N stands for a majority of the battery cycles, which may explain that when both FCR-N and daytime charging is prohibited, then the power tariffs barely increase, only by about €1 per month. However, this does not really matter since the revenue generated by these models (and in fact all models limiting FCR-N) is lower than the net revenue for all models which do not limit FCR-N, even after subtracting the increase in power tariff. Therefore, the FCR-N limiting models would have to actually decrease the power tariff quite significantly in order to compete with the non FCR-N limiting models. Based on these results, it can be concluded that setting hard limits on FCR-N, as done in these strategies, does not seem like a good decision in terms of battery profits.

5.7 Battery Cycles & Cycle Cost

The batteries analyzed are, as mentioned previously, rated for 10 years or 10,000 cycles, whichever comes first. In tables 4.2 and 4.5, it can be seen that none of the strategies analyzed is even close to using 900 in the studied period, which means that no strategy is close to 1,000 cycles per year or 10,000 in 10 years. This means that battery cycling does not seem to be a limiting factor at all, instead calendar aging is much more limiting. Based on this, it seems reasonable to have a cycle cost of 0.

There are a few reasons why the number of cycles is kept low. Firstly, the electricity tax is an extra cost which is added when buying and selling energy. Therefore, buying and selling electricity back to the grid becomes slightly less valuable, and thus the number of cycles used is decreased. Secondly, the efficiency of the battery makes it so that some energy is lost both when charging and when discharging. This means that if the battery is charged by buying 10 kWh of energy, then only about 8 kWh can be sold back to the grid, the rest is lost. Both of these factors mean that the value of the charging and discharging is lowered and that arbitrage is not worth it at all unless the price spread is large enough.

During research, it was found that having a battery rated for 10,000 cycles is quite rare. Instead, it seems more common for them to be rated somewhere in the low thousands of cycles. This means that the assumption of 0 cycle cost may not be suitable for the majority of batteries. For modeling where battery cycles are a limiting factor, something that could occur if the battery is rated for fewer cycles, market dynamics are different, or different markets are explored, a few ways of limiting the battery cycling have been identified. Firstly, one could change the cycle cost to be larger than zero and if necessary tune the cycle cost to make sure the battery will maximize profit during its lifetime. Alternatively, one could have a cycle cost of 0 but instead set a constraint in the model that the number of cycles must be below the what the battery is rated for, this can be done both for the entire lifetime of the battery but also for smaller time-periods by, for example, limiting to 1 000 cycles per year or 80 cycles per month.

5.8 Model Choice

Early on in this work, both a LP and a DP model was built. After a while, it was chosen that only the LP model was going to be finished and used as the final model. The reason for this is that the LP model was much simpler to work with and scale. LP works by adding constraints and an objective function to maximize, and the model then finds the best actions. DP instead works by creating the objective function and then adding all possible actions. This means that if the model is to be scaled, like adding a new market, then the LP model only needs the constraints and requirements for the specific market and one term in the objective function to be added. DP on the other hand, needs all of the allowed actions in the new market and all combinations of actions in the new and old markets to be found and added to the model, and these actions also need to be discretized in order to work with the model, which is significantly more cumbersome. This finding is in line with what other work found, that when using DP for this problem, one may struggle with the dimensionality. Similarly, if a market changes its requirements or market mechanics, one only has to change the relevant constraints if LP is used while if DP is used, then one has to recalculate all allowed combinations of actions across all different markets and add them to the model. Based on this reasoning, LP was deemed to better fulfill the goal of being an easy-to-scale and manipulate model compared to DP and therefore it was chosen to only continue to develop the model LP.

Chapter 6

Conclusions and Future work

The research successfully met its main goals by developing an optimal multi-market & arbitrage optimization model, evaluating price forecasting models, and demonstrating the combined performance in a real-world setting based on historical data.

This chapter will present the conclusions of the study as well as limitations and potential areas for future work. Lastly, a couple of reflections regarding ethical, societal, and environmental aspects are brought up.

6.1 Conclusions

6.1.1 BESS scheduling & multi-market optimization

The first goal of the thesis was to develop a BESS optimization model able to generate an optimal BESS schedule containing SoC-levels, frequency regulation market participation and bid sizes for every hour of the day. This was successfully achieved using a MILP model, which proved to be a better alternative than a DP approach. The model effectively incorporates BESS technical limitations, specific Swedish FCR market regulations for Limited Energy Resources, and battery degradation costs. Simulations with perfect foresight of market prices demonstrated that the proposed multi-market optimization strategy increases potential revenue (up to 34.8%) compared baseline single-market strategies, such as only participating in FCR-D.

Frequency regulation services, particularly FCR-D Down, were identified as the dominant revenue source, with energy arbitrage accounting for negligible returns during the 2024 study period.

The results and the optimal BESS schedule showcases multiple different combinations of bids of various sizes that are possible under the new requirements for LER that are given by Swedish TSO Svenska Kraftnät. Simply comparing FCR market prices side by side, and then choosing to participate in the market with the highest current price is therefore not sufficient in order to find the optimal BESS schedule. A simple approach like that has been used in previous studies, but one of the conclusions of this thesis can be considered the fact that optimal scheduling involves placing bids of different sizes in multiple markets simultaneously, which requires careful consideration of the technical requirements.

Another conclusion is that battery-cycle degradation can be substantially reduced without any major impact on total profits. This can be achieved by introducing cycle costs into the optimization objective. However, even with a cycle of zero, the studied BESS is more likely to reach EOL-calendar life before exceeding its maximum number of charging cycles. Thus, a zero-cost assumption for battery cycles could be considered reasonable, but this could vary depending on the type of BESS used and the number of cycles it is rated for.

6.1.2 Power tariffs

Regarding power tariffs, utilizing a residential BESS for the electricity price arbitrage included in the multi-market optimization strategy will likely lead to increased power tariffs. Table 4.6 shows that for an average villa, using the basic implementation of the multi-market optimization strategy will increase the power tariffs by more than €140 per year (using the common residential BESS specifications listed in the parameters). However, by limiting charging to night hours and restricting night-time charging speed to 50 %, the increase in power tariffs can be effectively mitigated with only a 1 % decrease in total revenue.

This is based on the current implementation of power tariffs by the Swedish electricity company Ellevio, where only half of the power peaks are considered during night hours. Other electricity providers may have different implementations of power tariffs and, therefore, the results may not be applicable to all residential BESS owners. The results also depend on the size and power capacity of the BESS, where residential BESSs with higher charging and discharge power will likely increase the power tariffs even more.

Apart from engaging in electricity price arbitrage, the results show that providing FCR-N regulation will also increase power tariffs for most households. This can be limited by restricting participation in the FCR-N market during night hours. However, the loss of FCR revenue by doing so is larger than the increase in power tariffs, suggesting that the addition of FCR-N participation in the multi-market optimization is still beneficial.

It should be noted that the impact on power tariffs is heavily dependent on individual customer consumption patterns, meaning some BESS owners might see high increases in their power tariffs, while some might see no increase. The numbers presented in the thesis are an average based on 48 randomly selected households.

6.1.3 FCR market price predictions

Price forecasting models, including ARIMA, Prophet and simpler baselines (B_{yest}^{avg}, B_{yest}, B_{D-2}), were developed in order to provide necessary inputs for optimization under realistic conditions, where market prices are not known before placing the bids.

B_{D-2} generally had the best predictive metrics (MSE, RMSE, MAE and MAPE) for most FCR markets. However, forecasting FCR market prices, especially FCR-D Up, proved challenging, with high MAPE errors observed across all models.

This suggests that accurate predictions of the FCR capacity compensation market prices are hard to achieve, and further research could explore what other predictive factors could be incorporated into the predictions. The markets also show various occurrences of price ”peaks”, which happen for largely unknown reasons. More research could focus on predicting these, although it is unclear whether that is possible in practice.

6.1.4 Evaluation with Imperfect Price Predictions

Even with imperfect price forecasts, the multi-market optimization approach achieved revenue gains (ranging from +3.7% to +12.0%) over the baseline FCR-D-only strategy, proving its viability in a real-world scenario, where FCR market prices are not known beforehand. The model developed in the thesis, combined with the price prediction models, could therefore be successfully implemented and used by BESS owners and aggregators looking to increase revenue from individual BESS systems or BESS-fleets. The increase in revenue is lower than in the case of perfect price predictions, which is expected.

Scheduling based on ARIMA forecasts resulted in the highest revenue among the imperfect forecast scenarios, despite B_{D-2} having better metrics for most markets.

6.1.5 Main contribution & Insights

The primary contribution of the research is a framework for optimizing residential BESS fleets across Swedish FCR markets and energy arbitrage, driven by price-forecasting of FCR market prices.

A key insight is the clear dominance of participating in FCR services compared to engaging in energy arbitrage in the Swedish market for the studied period. This might, however, change if FCR capacity compensation prices continue to drop.

Another important insight is that even without perfect price predictions, the multi-market optimization strategy proposed in the thesis can achieve significant gains over any basic single-market strategy, proving its practical viability.

The analysis also demonstrated that, on average, limiting BESS charging power during day hours significantly reduced the impact on power tariffs, without compromising overall revenue too much.

Lastly, the research shows that the optimal bidding behavior often involves simultaneous participation in multiple FCR markets (FCR-N, FCR-D Up, and FCR-D Down) at the same delivery hour, fully utilizing the capabilities of the BESS and leveraging the current requirements for Limited Energy Resources given by Svenska Kraftnät.

6.2 Limitations

The study has a couple of limitations, as outlined in Chapter 1.

Firstly, the study focused on a specific time period (Feb-Dec 2024) and the Swedish electricity market (SE3). The results may not directly translate to other regions or future market conditions and changed regulations. Most importantly, the results are highly dependent on market prices, especially for the FCR markets. If these continue to drop, as they have done for the last half of 2024, the results presented in this thesis might change significantly. However, the model presented in this thesis can still be used to retrieve the optimal BESS schedule, even when market prices vary.

Secondly, the exclusion of PV production means that the optimality for households with installed solar panels is not addressed. This is mainly due to the problems with fleet coordination that arise when aggregating multiple residential BESS with local electricity production for participation in frequency regulation.

The power tariff analysis relied on a specific implementation of power tariffs by Swedish electricity company Ellevio. This only accounts for one possible implementation of power tariffs, and other implementations might require different strategies to mitigate power tariff impact. Furthermore, the calculations on power tariffs are based on a limited number of consumption patterns from individual villas. As mentioned earlier, any increases in power tariffs will depend heavily on individual consumption patterns. Thus, the results can only be interpreted as a general broad approximation for the increase in power tariffs if customers do not change their consumption patterns and implement the multi-market & arbitrage optimization strategy.

When it comes to battery degradation, the study uses an approximation of the battery cycles used based on the total discharging power. Using this does not account for the fact that many charging cycles with a smaller DoD might not contribute to the same amount of battery degradation as one larger cycle, even if the total amount of charging/discharging is the same, as has been pointed out in several other studies in section 2.6.3. However, the finding that calendar aging is likely dominant for the modeled BESS mitigates this to some extent. Another simplification is that the difference in calendar aging incurred at different levels of SoC is not considered.

Another limitation is that the study only simulates one particular specification of a residential BESS. Different specifications will yield results other than those presented in the thesis. However, the specifications chosen in the study is one of the most common battery specifications for residential BESSs among Greenely customers.

The multi-market optimization strategy presented in the thesis only incorporates the FCR-N, FCR-D Down, and FCR-D Up markets. The main reason behind this being that they have a set of clear requirements for LER resources, which makes them suitable to model using a MILP approach. The requirements currently allow for participation in all three markets simultaneously (as long as certain requirements are met). Additionally, they have relatively low minimum bid sizes (0.1 MW), which reduces the number of BESSs that need to be aggregated in a VPP in order to participate.

6.3 Future work

Future work could investigate the inclusion of additional frequency regulation markets such as mFRR and aFFR, as well as participation in the first FCR auction. These markets come with their own set of challenges and limitations. One of the challenges being the fact that their auctions close before the spot prices for the upcoming day are released [17]. This would require forecasting spot prices to be able to use energy arbitrage in an optimization approach like the one presented in the thesis. Potential inter-dependencies between different markets would also have to be addressed.

Secondly, research could go into producing more advanced forecasting techniques for the different FCR market prices. However, as this thesis and other studies have shown, accurately forecasting FCR prices is challenging, and more sophisticated techniques like LSTM struggle to beat baseline predictions [45]. Further research could focus on finding new exogenous variables (like weather or grid load) that can be used to improve the accuracy of the predictions.

Testing the proposed framework in a real-world VPP environment with actual BESS fleets could also be of scientific value, as this could highlight any practical implementation challenges that have not been addressed in the thesis.

Future work could explore the possibility of having individual residential BESSs engage in local optimization activities such as increasing PV self-consumption or peak-shaving, while simultaneously being part of a VPP fleet submitting bids in the frequency regulation markets. This is somewhat explored in this work using custom power limiting strategies, however, more advanced strategies for limiting power tariffs increases or even actively decreasing the power tariffs could prove valuable. Managing a fleet that optimizes for both global revenue maximization and local optimizations for PV and power tariffs is a difficult challenge, as aggregators need to ensure that the requirements for participation in the FCR markets are met at all times. However, if proven possible, the potential savings and earnings using an intelligently scheduled residential BESS could be even higher.

6.4 Reflections

From an economic point of view, this research demonstrates a clear framework for increasing the economic viability of residential BESS investments. By optimizing participation across multiple revenue streams, BESS owners can achieve higher returns and aggregators can see more profit from their VPP fleets. This, in turn, can stimulate more investments in distributed energy storage.

The social and environmental aspects of the study are mainly related to the ability to integrate more resources into balancing services and frequency regulation markets. This contributes to overall grid reliability and stability, which is beneficial to all electricity consumers. Climate change demands actions to reduce carbon emissions, and as more intermittent renewable energy sources such as wind and solar are incorporated into the electric grid, fluctuations in energy production must be handled, which intelligent BESS systems can contribute to. This work aligns with United Nations (UN) SDGs numbers 7 (Affordable and clean energy), 11 (Sustainable cities and communities), and 13 (Climate action) [1].

Ethical considerations include data privacy for households participating in VPPs, as individual consumption patterns can potentially be used to infer when people are home, or leave their homes, etc. The algorithms should be designed to not exploit any market design flaws or technical loopholes, as this might lead to increased grid instability. Ensuring that optimization strategies do not lead to excessive battery cycling, which decreases the battery life-span, is also an ethical responsibility towards sustainable resource use.

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